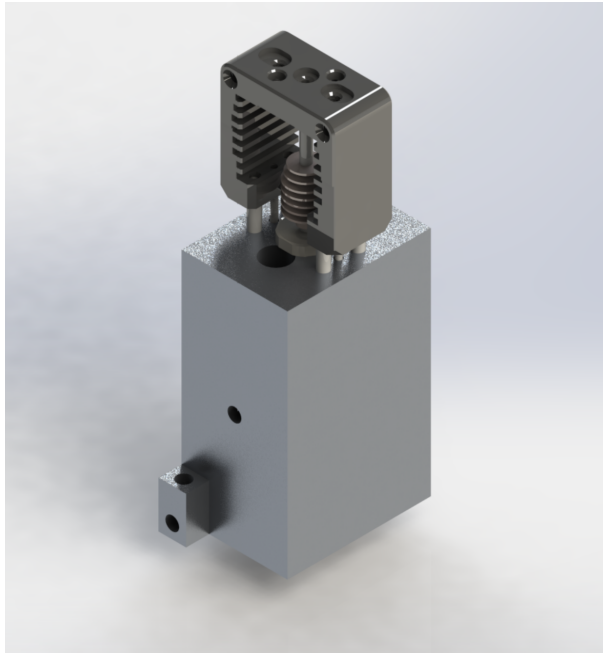


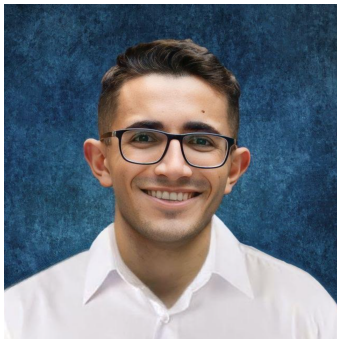
MEC4407

CONCEPT DESIGN SUBMISSION



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EXECUTIVE SUMMARY

The team was given a project from Scott Wordley, the CEO from 3D Printing Pty Ltd, a huge printing firm and under the supervision of our manager, Kathy, with the objective of developing a hotend for the Modix 3D printer which will allow prints at a faster flow rate. The team was asked to do an analysis and benchmark test of the Prusa MK3 i3 3D printer to be able to come up with a preliminary concept design that if the client finds suitable will be further tested and analysed to finalise a design. The types of analysis that were done was done by both CAE and hand calculations for structural and thermal aspects of the Prusa MK3 hotend. Benchmarking was done by utilizing a source of grey literature from Slice Engineering as a reference guide for different printing speed, temperature, and flow rates to test.

The structural CAE analysis found that the Prusa MK3 hotend has a safety factor of 14.7 in the X and Z direction and 14.9 in the Y direction under the the loading force of 40 N by applying the appropriate material for each component and that the possibility of screw failure in the hotend can be eliminated as a result of the high amount force of 18.6 kN needed to shear the stress.

The Thermal CAE and hand calculations analysis found that the Prusa MK3 hotend experiences a temperature ranging from 22°C up to just below 60°C in the heatsink and 220°C in the heat block, proving that the Prusa i3 MK3 hotend effectiveness in preventing heat creep, evident in the neck's significant temperature drop.

Benchmarking of the Prusa i3 MK3 hotend was done via two methods to determine the optimum maximum flow rate that the printer is operating at. It was found that the optimum flow rate ranges from the values of 10.2 mm³/s and 12 mm³/s at the printing temperature of 240°C and at speeds ranging from 240 and 250 mm/s.

The preliminary concept design that the team conceptualized is a hotend that utilizes the Super Volcano nozzle which has a longer nozzle that allows for the filament to have more time to be heated by the block and the ability to print at a greater flow rate which in return will be more sustainable in its operations as it uses less material and energy during its operation.

ACKNOWLEDGEMENT OF COUNTRY

Team 11 wishes to acknowledge the Bunurong people of the Kulin Nations, upon whose land they convene to engage in their concept design work. With deep reverence, Team 11 pays homage to the Ancestors and Elders, both past and present, who have lived on, nurtured, and cared for the Bunurong Country for tens of thousands of years.

There are many sites on Country that the Bunurong Ancestors interacted with. The Bunurong people historically dwelled in the green, lush lands between Mordialloc and Nunawading. Figure 1 depicts a part of the aboriginal cultural landscape: Karkarook Park (Aboriginal name for "Sandy Place" [1]), located near Monash University – the land at which Team 11 gathers. Karkarook Park stands as a recreational and environmental oasis, revegetated with indigenous plants. Similar to the land of Monash University, Karkarook Lake, located within the park, provided sustenance to the indigenous inhabitants and continues to do so, serving as a fishing spot for visitors. Team 11 recognises their past and ongoing role in caring for this beautiful Country.



Figure 1: Karkarook Lake in Karkarook Park [2]

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1.0 INTRODUCTION

The team was consulted by Scott Wordley, a CEO of a huge printing firm, to improve the design of the hotends attached to the company's Modix 3D printers. To do this, the team was asked to benchmark the E3D V6 hotend fitted on the Prusa i3 MK3 available at Monash University. Its design was also further refined, and adapted to a Slice Mosquito heat break while making sure of its compatibility with Bondtech BMG direct drive extruders.

This report covers the scope and timeline of the project at hand. The team has also worked to align with the United Nations Sustainable Development Goals, identifying a primary goal associated with the task. Through simulations and physical testing, benchmarks were created and presented in the sections below. These analyses consisted of hand calculations, structural Finite Element Analysis (FEA), and thermal analysis. Using these analyses and the group's ingenuity, creative solutions were generated. Of these solutions, two prominent concept designs of the hotend were formed. These concept designs were discussed and ranked accordingly, addressing the client's requirements and specifications. Furthermore, this report will later give advice for a plan for future work and preliminary conclusions.

1.1 BACKGROUND

Fused Deposition Modelling (FDM) is a material extrusion additive manufacturing technique where a thermoplastic such as polylactic acid (PLA) is melted by the hotend and squeezed out of the nozzle of the printer [3]. An example of an FDM printer is a Prusa i3 MK3.

The hotend is made up of three essential zones. Firstly, the cold zone prevents heat from spreading all over the hotend, potentially reaching the extruder, and melting the filament which causes blockage in the hotend. Secondly, the transition zone is where the hot and cold zone meets, and the solid to liquid change of state takes place. And lastly, the material reaches its molten state in the hot zone.

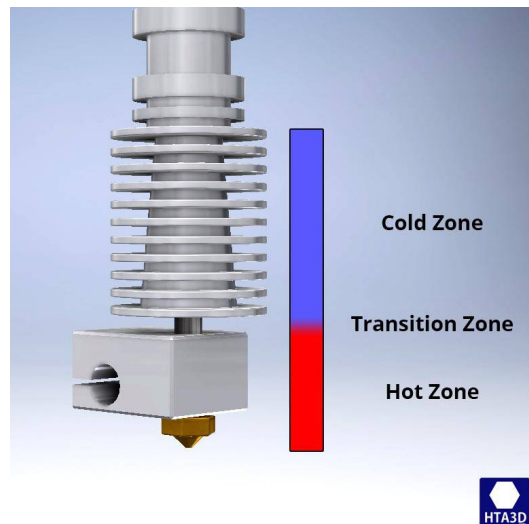


Figure 1.1.1: Hotend Zones [4]

There are six major elements that form the hotend: nozzle, heater block, heat sink, heat break, heating cartridge, and thermistor. The nozzle which is screwed into the heater block, is where the material is extruded from and focused onto the printer bed. The heater block is responsible for transferring heat to the nozzle and the heated half of the heat break. The heat break, which is screwed to the heat block and the heat sink, acts as a thermal break and hosts the transition zone as it directs the filament towards the nozzle and prevents premature melting of the material. The heat break is classified into two types: all metal, and plastic-lined where a Teflon insert is used to reduce friction. However, Teflon is not recommended above 265°C as it would start to break down [5]. Premature melting is known as heat creep. PLA has a glass temperature (temperature at which it starts to soften) of around 60°C and a melting temperature of around

180°C [6]. This glass temperature should start in the neck of the heat break and the melting temperature should occur through the heat block. If heat diffuses and the glass or melting temperature occur higher up in the heatsink, this is known as heat creep and it can affect the print drastically, often causing the PLA to anneal, resulting in a blockage [7]. The heat sink cools the heat break in order to ensure separation of both the cool zones and hot zones of the hotend. Also both connected to the heat block, the heat cartridge heats up the heat block while the thermistor measures the block's temperature.

The Prusa i3 MK3 in Monash University is installed with an E3D V6 Hotend, promoting an all metal design accompanied by high temperature performance [8]. The E3D V6 Hotend claims to have a high temperature performance reaching up to a temperature of 300°C - 500°C, by changing the components like the sensors from a thermistor to a Pt100 or Pt1000, and the heat block from aluminium to plated copper respectively. Additionally, its temperature sensors and heater cartridges can be easily replaced as they are just slid into the block and tightened by a grub screw. The hotend had a reduction in the extrusion force and an increase in the reliability of its retractions which increased its precision. This was all due to the all-metal heat break that maintained a sharp thermal break while highlighting an improved internal surface finish [9]. On the contrary, the hotend still vulnerable to issues which could affect its flow rate such as clogging of the nozzle from burnt plastic, or by using a faulty cooling fan and many more.

The open-source British manufacturer E3D made the parts of the V6 hotend compatible with a wide spread of products in the industry which made it possible to reuse and swap it with a wide variety of different models. This produced a number of combinations of new assembled hotends and also gave inspiration to imitators in the business. E3D Revo Six is one of them, where both its nozzle and heat break are fused together in a single screw which made its removal from the assembly effortless, and minimized the chances of filament leakage. However, it only works with a specific nozzle and only heats up to a temperature of 300°C. And, it approximately doubles the price of the V6 [10].

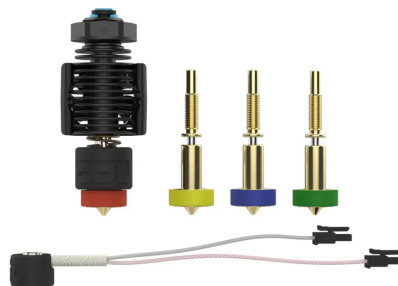


Figure 1.1.2: E3D Revo Six [10]

Slice Engineering also made their version of a hotend that competes with the V6 called the Mosquito. Its notable bimetallic heat break made of copper alloy instead of aluminium, allowed a 85% decrease in heat conduction up the filament path preventing heat creep. Other than this, the aircraft grade aluminium heat sink is built with a roll cage design that protects the heat break from abuse. Similar to the V6, its temperature is rated up to 500°C. On the downside, the price of the Mosquito triples that of a V6 [11].



Figure 1.1.3: Slice Engineering Mosquito Hotend [11]

E3D further improved the design of its hotend by introducing the SuperVolcano. It is similar to the V6 but produces a significantly high flow rate, believed to be eleven times greater [12]. A long hot zone is produced by utilizing a supervolcano nozzle, a bigger heat block, and a more powerful heat cartridge. This allows the filament to get heated longer and permits a much smoother print as it extrudes out of the nozzle. As the heat cartridge is much stronger, it is likely to melt an aluminium block, therefore the heat block of the supervolcano is a nickel plated copper alloy while also incorporating a high temperature fiberglass sleeve for insulation [12]. Though expensive, it is relatively cheaper compared to Slice Engineering Mosquito [10].



Figure 1.1.4: E3D SuperVolcano Hotend [13]

E3D hotends are not restricted to single filament printing, there are also models that are capable of generating multi-filament prints just like the E3D Chimera+ which works similar to the E3D Cyclops+ [10]. The E3D Chimera+ makes an assembly of two hotends which enables two different filaments to flow through. This results in dual colored prints making the prints much more appealing to the eyes. But, it is prone to leakage from one of its nozzles if only one of the extruder is operated. Thus, both extruders must be used simultaneously. By using multi-material filaments in a print, water-soluble supports can be exploited as well.

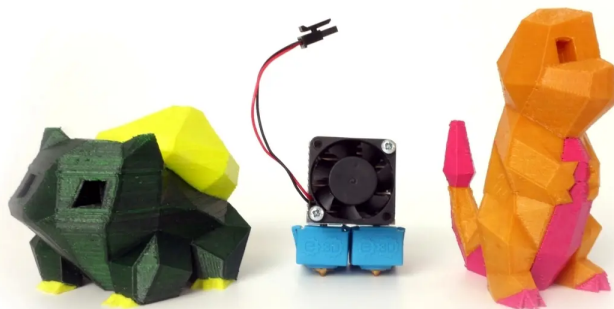


Figure 1.1.5: E3D Chimera+ Hotend [10]

The most commonly used nozzles in the market, screwed into the hotends mentioned above are as follows: an E3D Standard, Volcano and SuperVolcano with an approximate thread length of 6 mm, 16 mm, and 45 mm respectively. Moreover, a Standard Slice Engineering Nozzle with a thread length of around 7mm. Out of the four nozzles, the SuperVolcano had better flow rate mainly due to its length.

1.2 LITERATURE REVIEW

Slice engineering tested the maximum practical flow rate of its Mosquito Magnum+ in the FDM printer without compromising the quality of the prints. The Average volumetric flow rate was measured by producing a set of test prints operating at different speeds using a 1.75 mm diameter filament. Also, different baseline speeds were used for each model [14].

The test models were 20 mm tall and had a specific geometry to illustrate the adhesion between the layers. Three different industry popular filament materials were inserted into the printer: Polylactic Acid (PLA), Acrylonitrile Butadiene Styrene (ABS), and High Impact Polystyrene (HIPS).

As the test started, the speed of the printer was increased by 5 mm/s for every 5 mm interval of the 20 mm height of the model. Whereas, two temperatures were used: according to the manufacturer's recommendation and boosting it up by 60°C. The average volumetric flow rate was calculated by measuring the diameter of the filament and its length, both in millimeters, and print time in seconds. These values were then inserted into Equation 1.

$$Q = \frac{\pi d^2 l}{4t} \quad (1)$$

The maximum volumetric flow rate where defects started to show was also tested, and defects found out to exist only at the last two test prints for each filament. It was observed that printing ABS at 240°C generated the maximum flow rate of 53 mm³/s and 91 mm³/s at 300°C. For PLA, it was 65 mm³/s at 220°C and 88 mm³/s at 280°C. And for HIPS, the value was 71 mm³/s and 93 mm³/s at 245°C and 305°C respectively.

Another way of benchmarking a hotend of the 3D printer is by weighing the print to check its quality, as done by CNC Kitchen. A round structure is printed using a Prusa i3 MK3s paired with an E3D V6 hotend. The temperature was kept at 215°C and the flow rate was increased starting from 3 mm³/s on a 3 mm³/s interval to get a variety of prints. Each test print was then weighed. Then, a graph of weight against volumetric flow rate was plotted that formed a trend that illustrates the decrease in weight when the flow rate was increased [15]. This was due to the extruder skipping at higher flow rate which caused slips in the extrusion system, where less material was getting extruded from what is expected.

However, after doing an extrusion test where the temperature was increased up to 275°C and fixing the flowrate at 15 mm³/s, slipping decreased to the point that it was non-existent at a temperature of 275°C. Thus, showing at a reasonably higher temperature, the quality of the print improves. This trend occurs as at higher temperature, the viscosity of the melted plastic decreases putting less pressure in the extrusion system [15].

1.3 REQUIREMENTS AND SPECS

The requirements and specifications of the design of the hotend is given in Table 1.3.1.

Table 1.3.1: Requirements and Specifications

Requirements	Criteria	Weighting (/10)	Specification
Performance Hotend Flow Rate	Flow Rate of the filament out of the nozzle in cubic millimeter per second	10	20% - 50% increase
Performance Hotend Temperature	Appropriate temperature to produce good quality prints while keeping the flow rate at optimal level	8	230°C -240°C
Performance Layer Height	Height of the printed layer in millimeter	7	80% of the nozzle orifice ≤ 0.32 millimeters
Performance Layer Width	Width of the printed layer in millimeter	7	150% of the nozzle orifice, ≤ 0.6 millimeters
Aesthetic	Focus-group judged	3	N/A
Safety	Prevents Heat Creep, heatsink does not burn the user when touched	6	Must comply with Australian Standards (AS) safety guidelines.
Cost Affordable	Manufacturing cost in Australian Dollar	3	ROI of 5-10 years of the project cost.

2.0 PROJECT SCOPE AND TIMELINE

In order to address the client's needs to improve the design of the hotends attached to the Modix 3D printers, the process has been broken down into two overarching deliverables, a preliminary concept design report and presentation; and a final design report and presentation. The goal of the preliminary concept design report is to develop and indicate the necessary understanding and skills needed to create an improved hotend and assess its performance. If the client is happy with the team's fundamental skills and analysis methodology that satisfies both their budget and contractual needs, then a final design report will be conducted, whereby the goal of the final design report is to perform rapid prototyping to ultimately manufacture an improved hotend for the client's Modix 3D printer - validated via theoretical hand and computer simulated calculations as well as physical testing.

Regarding the preliminary design, the client would like the standard hotend of a Prusa Mk3 i3, the E3D V6, to be benchmarked for performance both analytically and experimentally. Analytically, the structural and thermal integrity of the E3D V6 hotend is investigated, while experimentally, physical testing including 3D printing and utilising a thermal heat gun should be conducted. To do so, the team has broken up into three sub-teams, including thermal analysis, structural analysis, physical testing, to finalise the benchmarking in a shorter amount of time as there is little dependency between each analysis. For the thermal analysis, the two major deliverables include performing hand calculations and computed aided engineering simulations to determine the heat break performance at 230°C as well as the flow rate performance at maximum heating rate of 40W. In order to do so, relevant tasks included identifying the relevant conditions and assumptions, finding the material properties required and performing relevant hand calculations to determine the theoretical values. This is followed by determining the relevant conditions and their assumptions for the simulation model, determining an appropriate mesh, inputting the correct conditions and calculating the simulation values, which should be similar but may vary slightly due to different assumptions made. For the structural analysis, a similar approach is applied, with tasks including identifying the relevant structural challenges and assumptions, finding the necessary material properties required and performing the relevant hand calculations to determine the theoretical structural challenges. This is also followed by determining the relevant boundary and loading conditions and assumptions used for the simulation model, performing a convergence check to validate the mesh and then performing the simulation analysis to determine the simulation calculations. Meanwhile, for physical testing, the appropriate tasks included researching academic literature to help produce a standardised 3D printing test, follow the guides from academic literature to test and determine the flowrate and speed - which is an iterative process. This is then followed by thermal heat gun analysis, which requires the E3D V6 heat block to be painted black to prevent any infrared results interference, heating the hotend and then capturing the thermal diffusion of it.

After completing some proper performance testing on the commercially available option that is on-hand, the client would like a solution generation and a final preliminary concept suggestion. The client would like the team to design a final preliminary concept of a heat block that can hold two 60W heater cartridges (Diameter 6mm and length 50mm), a PT1000 thermistor (diameter 3mm and length 15mm) and a off-the-shelf nozzle (with thread lengths of either 7mm, 16mm or 45mm). This design should be easily integrated with a Slice Mosquito heat break and heat sink using the four standoff tubes, as well as a BondTech BMG direct drive extruder and either a GDA010 Standard Slice Heat Break Fan or a N052AR Heat Break Fan. To address this, the relevant tasks included considering current factors noted from current benchmarking analysis and understanding the client requirements, determining on a decision making criteria to assess the future solutions, developing a morphology table, generating multiple design solutions using that table, choosing two solutions of the best solutions to go into more detail, evaluating the solutions against the criteria previously established and determining the final preliminary concept design while also considering the statistical significance. The decision making criteria is done before the creative solution generation to help assist in the development and choosing of creative but appropriate solutions.

The team will then identify their research and findings in a report, followed by a 10 minute presentation to the client. If a contract is formed and the client wishes to proceed to the next stage, a rough outline of the scope would be to ultimately manufacture a hotend for the client's Modix 3D printer that has an improved flow rate. This would involve a range of tasks, including adopting feedback from the client to determine to finalise on a design, finalising on a material selection, performing structural and thermal hand calculations and simulations to help evaluate the design, creating a detailed drawing for manufacturing, getting it professionally manufactured, conducting physical performance tests (including 3D printing and thermal heat gun), making any necessary modifications and repeating these steps one more time to produce a final hotend design and produce for the client. It is important to note that during this prototyping phase, testing should be conducted on the Creality CR10 S5 printers available even though a Modix 3D printer is the final end use case.

An overview of the entire project and the timeline is summarised in the Gantt Chart that can be seen in Figure 2.0.1.

3.0 UNITED NATIONS SUSTAINABLE DEVELOPMENT GOALS

The project's objective is to create a hotend that has a faster flow rate which in return would create a product in faster time and uses fewer materials and energy. The appropriate United Nations (UN) Sustainable Development Goal (SDG) for the objective that is stated above is UN SDG 9 [16] which is to build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation and can be seen in the Figure 3.0.1.

UN SDG 9 has two targets and subsequent indicators that are in line with our objective.

The first is target 9.2 [17] which describes the goal to promote inclusive and sustainable industrialization. This is in line with our objective to create a hotend that prints faster which ultimately makes it more sustainable in the usage of its material as it uses less material to create a print. The indicator 9.2.1 [17] which discusses manufacturing value is fulfilled by the project's objective which has the outcome of more products printed in less time using less material and energy that increases the manufacturing value of operating the desired hotend.

The second is target 9.4 [17] which has the goal to upgrade infrastructure and retrofit industries to make them sustainable with an increase in efficiency of material usage and greater adoption of clean and environmentally sound technologies. This aligns with our objective of creating a hotend that has a faster flow rate, as the result of that is that we use less material and energy during the operation of the hotend, by rigorous benchmark testing the printer to find the maximum efficient flow rate that it operates at and utilizing nozzle choices that suit this criterion. The indicator 9.4.1 [17] which states that the amount of CO₂ emission per unit volume is noted is fulfilled by the project's objective which has the result of less energy and material consumption that creates less CO₂ emission being created by the hotend during operation.



Figure 3.0.1: UN SDG 9 Goals [16]

4.0 BENCHMARKING: CAE ANALYSIS

Computed-aided engineering (CAE) analysis was performed to determine the structural integrity and the theoretical thermal characteristics of the Prusa Mk3 i3 V6 hotend. Hand calculations were initially performed to help validate and support the more thorough analysis performed on the computed-aided software package, ANSYS.

4.1 STRUCTURAL INTEGRITY ANALYSIS

Structural integrity analysis was performed to determine how the current hotend design will yield, given different point loads. Hand calculations were performed to help support the Finite Element Analysis conducted on ANSYS.

4.1.1 HAND CALCULATIONS

Hand calculations of suspected material failures allow for a general understanding and magnitude of what type of forces are acting on the body. Because these are rough calculations, they are not imperative conclusions of where the hotend will fail during operation. The CAE analysis of the system is almost always more likely to yield the most accurate theoretical results, and thus is more critical.

The greatest contact force the hotend will encounter is commonly a collision with a printed part on the printing bed. Thereby, a hand calculation of the potential collision can be modelled dynamically as a change in momentum of the whole hotend. While also keeping in consideration the different maximum velocities in each direction will yield two different resulting collision forces. One force in the z direction, as the hotend travels at 12mm/s, and another in both the x and y direction, as it travels at a maximum of 200mm/s. Therefore, from Equation (2), Equation (3) and given maximum velocities in each direction, resultant forces can be calculated from potential collisions [18].

$$\text{Impulse} = \text{mass of the system} * (\text{maximum velocity} - \text{end velocity}) \quad (2)$$

$$\text{Resultant force} = \text{impulse} / \text{time of collision}, \text{ Assume time of collision as } 0.1\text{s} \quad (3)$$

Therefore, the potential collision force in the z direction resulted in 0.0043N and 0.072N in the x and y direction. Results from Equation (2) and Equation (3) can conclude that the potential collision forces are significantly low in magnitude, thus material failure is impossible to occur in this scenario.

A more complex calculation can be explored around the possibility of threading failure in the hotend's screws. Specifically the nozzle screwed into the heat block, and the heat break screwed into the heat block and heat sink. In order to simplify the failure test, only the thread stripping of the heat break needs to be made, as the tensile strength of steel is lower than brass, which is the nozzle's material. Therefore, the minimum force needed for threading failure will be in the heat break before the nozzle.

To conduct thread stripping calculations, the threading form of the screws must be known. Further research into the industry standards and the specific detailed drawings of the E3D hotend heat break allows a rough magnitude to be determined. By first calculating thread length using Equation (4), then calculating the shear area of each thread using Equation (5). Thus the resultant force to shear the threads can be determined through Equation (6) and the yield stress properties of S_y .

$$t = 1.5 * d \quad (4)$$

$$\text{Shear area} = \pi d * 0.75t \quad (5)$$

$$\text{Force required to shear threads} = \text{shear area} * 0.58S_y \quad (6)$$

The force required to shear the threads in the stainless steel heat break is calculated to be 18.6KN. Due to the high magnitude in force required for screw failure in the hotend, the possibility of screw failure can be eliminated. All hand calculations remaining are listed in the [Appendix B: Structural Hand Calculation Analysis](#) that outline all results. From here, further analysis can be conducted electronically with simulated forces.

4.1.2 FINITE ELEMENT ANALYSIS

4.1.2.1 OBJECTIVES

Obtain results for deformation, shear and Von Mises stress to deduce a benchmark of the Prusa MK3 i3 hotends to compare at a later stage with concept hotends. By comparing to a baseline, we will more readily be able to quantify improvements made by our design.

4.1.2.2 ASSUMPTIONS

Through visual inspection of some of the Prusa printers in the studio, a fixed condition was applied around the face of the heat break towards the top where it would secure to the extruder. From research, the nozzle is assumed to be brass C37700, heating block aluminium 6061, heat break stainless steel 316 and heat sink aluminium 6061 [20].

4.1.2.3 LOAD CASES

A rather arbitrary force of 40N is selected as smaller forces such as those referenced in hand calculations do not demonstrate the desired resultant forces to a great degree. Forces or 'hits' experienced by the printer are assumed to be in the X, Y and Z directions and applied at the nozzle. The Y hit is applied directly vertical to the downward facing flat face of the nozzle as this is expected to collide first based on visual inspection. The X and Z forces are applied to one of the faces of the hexagonal component. These faces were selected rather than the conical component as these both protrude more and thus would collide first, and secondly provide a perpendicular surface to apply the force to so as not to encounter any analytical errors such as what may occur attempting to apply the force in vector components on the sloped conical face.

4.1.2.4 ANSYS

Finite element analysis (FEA) was conducted on the current printer hotend model to benchmark and compare results once an improved hotend is developed. CAD files were sourced online from resources including GrabCad and official websites of Slice Engineering and E3D. Online sourced models were verified with sample parts available in the design and build studios. This involved cross-referencing dimensions and visual inspection of the overall part. CAD specific verification was then undertaken within ANSYS discovery utilising the inbuilt repair tool to locate any parts or surfaces that may result in errors during the analysis.

To simulate forces that may be experienced during normal operation of the 3D printers. "Hits" were simulated by applying forces to the nozzle in X,Y and Z directions with a force of 40N.

Materials selected are constant across all simulations. Stainless steel 316 applied to heat break, aluminium alloy 6061 applied to heating block and heat sink, brass C3770 applied to nozzle. These material selections are based on market research of most common materials for relevant parts. [20]

The geometry is imported to a project within a static structural analysis in ANSYS workbench. The part is meshed checking for quality with aggressive mechanical limits with a target element quality of 5e-2 which is the default setting applied by ANSYS.

Results for Deformation are checked referring to mesh convergence to verify results. Stress will inherently not converge.

4.1.2.5 FEA RESULTS

Simulating the X hit on the nozzle, a force of 40 N was applied to one flat face in the X direction. By applying the force here instead of at the tip of the nozzle, the force applied is spread over a greater area to reduce abnormal results from ANSYS analysing a force applied to a minute area. The face selected can also reasonably be assumed to experience hits whilst in operation.

At a force of 40N, maximum deformation was found to be 1.2668e-004 m occurring at the point of applied load on the nozzle.

Results from safety factor analysis seen in Table 4.1.2.5.1 indicate average safety factor of the hotend at 14.709, with a maximum of 15 however this is the greatest result that is displayed by ANSYS. Minimum safety factor indicated was 1.489.

Table 4.1.2.5.1: Safety Factor for X hit at 40N

<i>Time [s]</i>	<i>Minimum</i>	<i>Maximum</i>	<i>Average</i>
1	1.489	15	14.709

These results shown in table 4.1.2.5.1 indicate that overall, the current hotend is relatively robust against an assumed average force of 40N hits in its X direction experienced whilst in operation. A low minimum value below a recommended SF of 4 could be assumed to occur as a result of the nozzle being the contact for the applied force and also having a small surface area. In operation, the nozzle is prone to experiencing chips and breakages which is why it should be easily swapped out.

Table 4.1.2.5.2: FEA Results X Hit 40N

<i>X at 40N</i>	<i>Total Deformation (m)</i>	<i>Shear Stress (Pa)</i>	<i>Von Mises Stress (Pa)</i>
Minimum	0	-4.902e+7	913.25
Maximum	1.2668e-4	5.9839e+7	1.6931e+8
Average	4.7241e-5	3.2507e+5	4.335e+5

Further analysis of forces in the Y component exhibit similar characteristics. Deformation is marginal across the entire body visualised in [Appendix C: Structural CAE Analysis](#) Figure C.2 and is maximum at the point of applied force. This is similar for shear and equivalent von mises stress as seen in table 4.1.2.5.3.

Table 4.1.2.5.3: FEA Results from Y Hit

<i>Y at 40N</i>	<i>Total Deformation (m)</i>	<i>Shear Stress (Pa)</i>	<i>Von Mises (Pa)</i>
Minimum	0	-1.1886e+7	191.87
Maximum	1.4101e-6	1.7523e+7	5.68e+007
Average	4.4249e-7	-3607.5	9.9013e5

Table 4.1.2.5.4: Safety Factor for Y hit 40N

<i>Time [s]</i>	<i>Minimum</i>	<i>Maximum</i>	<i>Average</i>
1	6.4684	15.	14.956

Results from the Y hit indicate the Prusa hotend is quite robust from collisions in this direction, mainly due to lack of an incurred bending moment unlike the horizontal hits from X which do create a bending moment, inducing stress particularly around the heat break.

Likewise with previous testing, deformation is again marginal. However, side hits such as that of X and Z exhibit higher equivalent stress and shear stress at the heat break. Furthermore, side hits produce higher stresses shown within Table 4.1.2.5.5 and Table 4.1.2.5.2 within the hotend across the board compared to that of vertical Y hits indicated in table 4.1.2.5.3.

Table 4.1.2.5.5: FEA results from Z hit

<i>Z at 40N</i>	<i>Total Deformation</i>	<i>Shear Stress</i>	<i>Von Mises Stress</i>
Minimum	0	-3.664e+7	201.75
Maximum	1.2604e-4	4.147e+7	1.9077e+8
Average	4.4275e-5	27299	4.3437e+6

Table 4.1.2.5.6: Safety Factor Z hit 40N

<i>Time [s]</i>	<i>Minimum</i>	<i>Maximum</i>	<i>Average</i>
1	1.335	15.	14.7

As is seen in Table 4.1.2.5.1 and Table 4.1.2.5.6, X and Z hits exhibit similar safety factor results whilst Y exhibits less stress and higher minimum safety factors overall. Furthermore, across both X and Z hits, deformation is greatest at the nozzle illustrated in [Appendix C: Structural CAE Analysis](#) Figures C.1 and C.2 whilst shear stress and von mises stress are greatest at the heat break between the heating block and the heatsink which is expected as the top of the heatsink is a fixed condition along with this section being smaller in radius compared to the block and heat sink.

4.2 THERMAL ANALYSIS

The following section outlines the thermal analysis performed. For a more detailed explanation and a link to the spreadsheet used to perform the hand calculations, please refer to [Appendix D: Thermal Analysis](#).

4.2.1 OBJECTIVES

To develop a better understanding of a current competitor's hotend, thermal computed-aided engineering (CAE) simulation results were analysed for the Prusa i3 MK3 printer's E3D V6 hotend design. The objective of the thermal analysis is to assess the performance of the E3D V6 heat break and calculate the maximum flow rate of the hotend. It is important to assess the performance of the heat break as it will indicate whether heat creep occurs meanwhile it is as important to analyse the maximum potential flow rate of the hotend to determine if the flowrate is limited by the power or whether it is due to the design of the hotend.

4.2.2 ASSUMPTIONS

Hand calculations are necessary to indicate the validity of the CAE simulation results. However, the results may vary slightly due to the many necessary assumptions made in the hand calculations to streamline the complexity of the analysis. Some foundational assumptions include assuming steady-state heat transfer occurring exclusively in one direction along the z-axis; maintaining constant thermal material properties and an ambient air temperature of 22°C; considering negligible radiation and contact resistance; assuming constant forced convection throughout the heatsink; accounting for constant natural convection on the top of the heat block and at the bottom of the nozzle, while assuming negligible natural convection elsewhere (i.e., along the sides of the heat block and hotend); assuming that overlapping components are at the same temperature; considering the maximum airflow rate of the fan as the actual airflow rate; assuming that constant high temperatures do not adversely affect the materials of the hotend; and considering discretizing the heat break five times, while discretizing the heatsink for every fin and gap as an appropriate method. To facilitate the modeling process, the analysis considered only four major components out of the total 11, with a simplified approach to modeling the nozzle, heat break, and heat sink as hollow cylinders with varying outer diameters, and the heat block as a rectangular prism with a single cut extrude for the heat break and heat sink.

4.2.3 MATERIAL CHOICE

For the four essential parts of the hotend, the thermal conductivity was determined by extracting the values from ANSYS GRANTA Materials Data for Simulation table (in ANSYS WorkBench) and the values are summarised in Table 4.2.3.1. It is important to note that these values are at a temperature of 300 K.

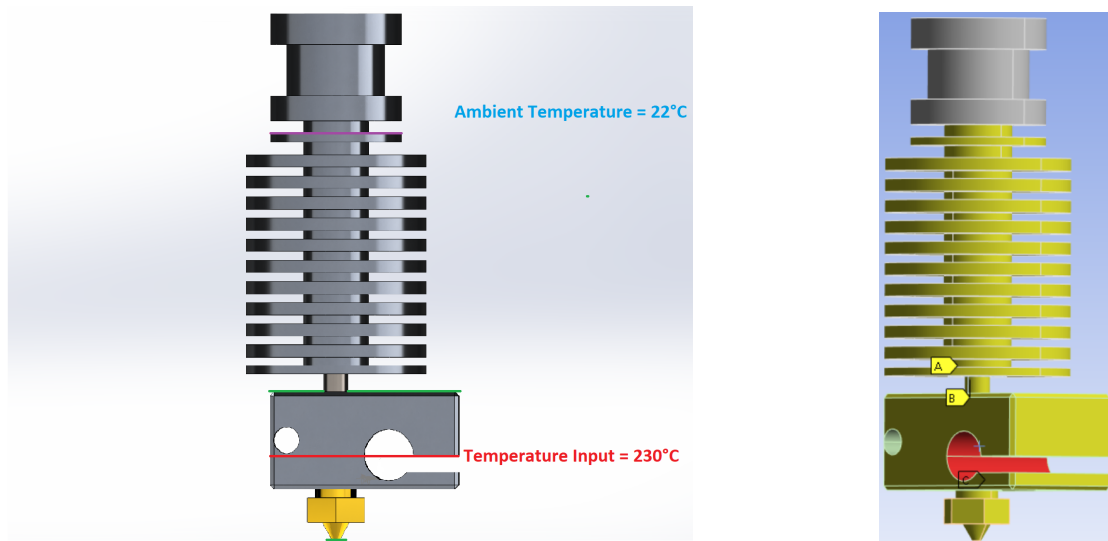
Table 4.2.3.1: Material Properties of the Four Major Components of the Hotend at 300 K [21]

<i>Component</i>	<i>Material</i>	<i>Type</i>	<i>Thermal Conductivity (W/mK)</i>
Nozzle	Brass	C37700	112.1
Heat Block	Aluminium Alloy	6061, T6	155.3
Heat break	Stainless steel	316	14.58
Heatsink	Aluminium Alloy, 6061	6061, T6	155.3

4.2.4 LOAD CASES

For both the hand and CAE analyses, the input temperature is set to 230°C, while the ambient temperature is set to 22°C, and the load cases can be seen in Figure 4.2.4.1. In hand calculations (the Figure on the left), the temperature is input at the center of the heater cartridge (red line), and the ambient air interacts at

the end of the nozzle (green), the top of the heat block (green), and the top of the heat sink (purple). Meanwhile, in the ANSYS analysis, forced convection with the ambient air occurs in the heat sink (A), natural convection occurs throughout the entire hotend (B), and the input temperature is located along the curved surfaces where the heater cartridge sits (C).



(a) Hand Calculations Load Case

(b) ANSYS Load Case

Figure 4.2.4.1: The Load Cases Applied for Thermal Analysis

4.2.5 METHODOLOGY

Prior to evaluating heat break and flow rate performance, hand calculations are necessary to derive natural and forced convection heat transfer coefficients. These coefficients are determined using the Continuity Equation for the air speed along the fins, Prandtl and Reynolds numbers for the Nusselt equation. Notably, film temperature is essential for calculating air properties in forced convection. Each component is discretized: the heat block into 3 parts, the heat break neck into 5, and the heat sink into 24. Individual resistances are calculated and overlapping components are treated in parallel. The resulting resistance is computed by summing these parallel equivalent resistances in series. This allows the heat loss at the top and bottom to then be calculated, resulting in the temperature throughout the hotend to be easily determined. For the CAE simulation, natural and forced heat transfer coefficients are applied to their respective locations, whereby natural convection along the side is not neglected. Various mesh qualities are tested, leading to the selection of an appropriate final mesh.

4.2.6 RESULTS

A summary of the key temperatures from the hand calculations can be found in Table 4.2.6.1.

Table 4.2.6.1: Hand Calculation Values at Important Points in the Hotend

Bottom of Nozzle Temperature (°C)	Bottom of Heat Block Temperature (°C)	Top of Heat Block Temperature (°C)	Bottom of Heat Sink Temperature (°C)	Top of Heat Sink Temperature (°C)
229.99	229.997	229.54	58.27	22.00

Additionally, a comparison of the amount of heat loss predicted from hand calculations and CAE simulation calculations through the top and bottom of the hotend is shown in Table 4.2.6.2.

Table 4.2.6.2: Comparison of Heat Out Calculations

Calculation Method	Total Heat Out through Heat Block (W)	Total Heat Out through Heat Sink (W)
Hand Calculations	0.0406	3.4174
ANSYS Calculations	1.6328	3.3726

Finally, the thermal CAE analysis can be seen in Figure 4.2.6.1, while the heat break performance and the flowrate performance can be found in Figure 4.2.6.2 and Figure 4.2.6.3, respectively.

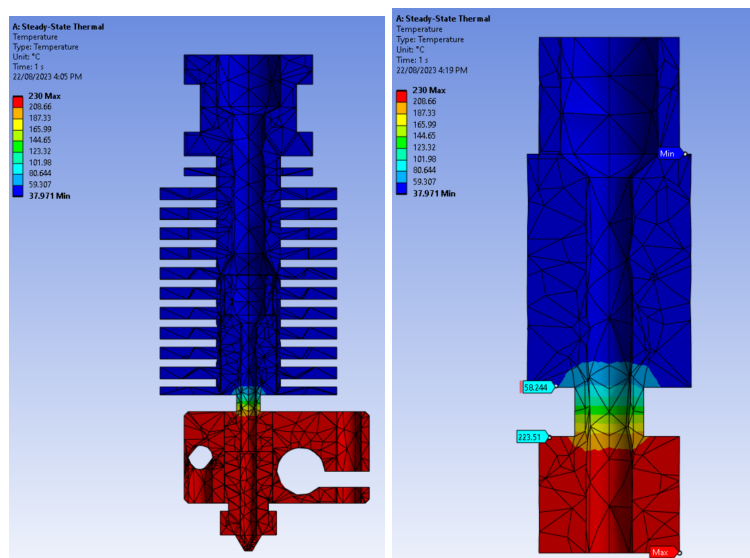


Figure 4.2.6.1: Section View of the CAE Analysis of the Entire Hotend and the Heat Break Itself.

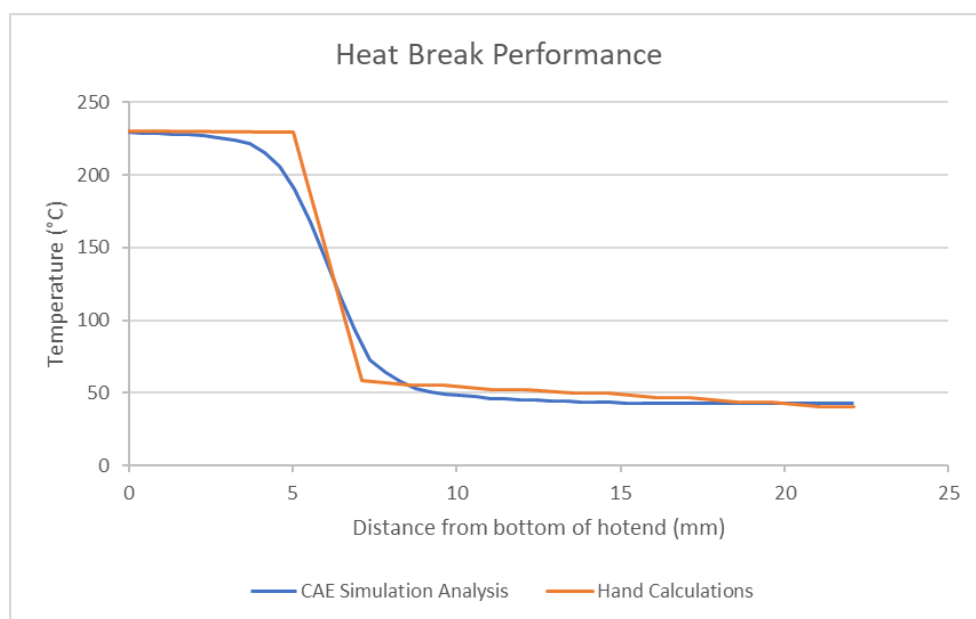


Figure 4.2.6.2: The Performance of the Heat break Via Hand Calculations

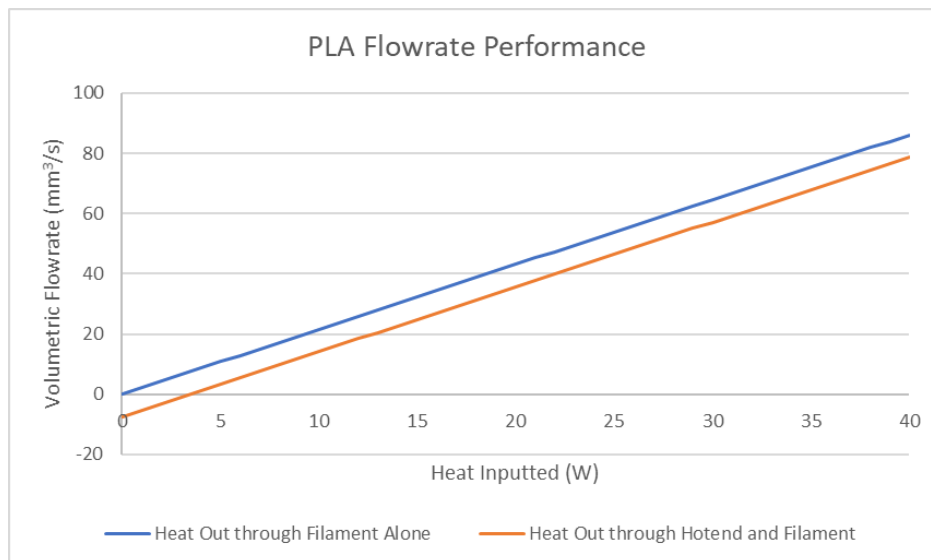


Figure 4.2.6.3: The Performance of the Theoretical PLA Flow Rates Possible via Hand Calculations

4.2.8 FINDINGS

The E3D V6 hotend of the Prusa i3 MK3 printer underwent a thorough analysis through both hand calculations and CAE simulation using ANSYS. As depicted in the temperature contour plot in Figure 4.2.6.1, the CAE analysis revealed that the majority of heat within the E3D V6 was concentrated in the heat block, with the highest temperature of 230°C occurring at the heat cartridge. Meanwhile, the heat sink exhibited temperatures mostly below the glass temperature of 60°C, with a minimum temperature of 37.971°C. Additionally, the CAE analysis indicated a temperature of 58.244°C at the bottom of the heat break in the heatsink, and 223.5°C at the top of the heat block. This suggests that the material begins to soften in the heat sink's bottom and melt in the heat block's neck, aligning with expectations. These findings are reasonably consistent with the results from hand calculations, where the bottom of the heat block measured 58.27°C and the top of the heat block was at 229.54°C as outlined in Table 4.2.6.1.

Table 4.2.6.2 displays comparable total heat output through the heat sink between hand calculations (3.4174 W) and simulation (3.3726 W). Conversely, total heat output through the heat block slightly differed, being 0.0406 W in hand calculations and 1.6328 W in the ANSYS analysis. Such discrepancies are most likely due to assumptions like 1D heat transfer and negligible natural convection along the sides and bottom of the heat block. Nonetheless, these minor deviations do not overshadow the overall consistency that bolsters the reliability of the CAE simulation results.

Figure 4.2.6.2 illustrates the performance of the heat break, highlighting a sharp gradient in the neck that signifies an effective heat break, preventing heat creep. The heat block and heat sink show a shallower gradient, indicating their proficiency in maintaining distinct hot and cold temperatures. Overall, the objective of assessing the E3D V6 heat break's performance was achieved, confirming its effectiveness in preventing heat creep, especially evident in the neck's significant temperature drop. Lastly, the potential flow rates under varying heat inputs of the hotend were analyzed and depicted in Figure 4.2.6.3. Applying a maximum heating rate of 40W, the maximum PLA flow rate was estimated at 78.71 mm³/s, assuming constant power through the bottom and top. Given E3D's claimed flow rate of ~15 mm³/s, this suggests that the flowrate is constrained by the hotend's design rather than its power. Thus, potential improvements in flow rate could arise through design modifications.

5.0 BENCHMARKING: PHYSICAL TESTING

Physical testing was done as a way to benchmark the E3D V6 Hotend using the Prusa i3 MK3 3D Printer. A source of grey literature from Slice Engineering [14] was used as a guide to benchmark test the hotend and used as an initial starting point on what parameters and variables that can be changed. An STL file was also sourced from Slice Engineering [14] that was used as the object benchmark that was to be printed. The STL file chosen was a model that is designed to test the 0.4 mm nozzle on a hotend. The reasoning on choosing this hotend is because the Prusa i3 MK3 3D Printer uses a 0.4 mm nozzle and would give the most accurate data. The objective of benchmarking the hotend is to determine the optimum parameters to achieve the maximum flow rate of the hotend and compare the experimental results with theoretical values.

We employed two different methods of test in benchmarking the Hotend. The first method is to print a scale of the object with a height that is set to 5 mm instead of the full 20 mm and to print and time the printing process until it is fully printed. The second method is to print the object and stop the printing when it's 600 seconds using the full object as sourced from Slice Engineering [14]. Both methods utilized Ultimaker Cura to set the parameters or settings of the print.

In addition to the two methods, a thermal gun is used to measure the temperature of the hotend during the printing process to see how much heat is each component experiencing.

5.1 METHOD 1: PRINTING A SCALED MODEL

The first method uses a scaled model of the object where its height is set to 5mm and then time the time it took to finish the print. It also uses the same types of parameters that were used in the first method which were layer height, layer width, Bed Temperature, Printing Temperature and Printing Speed. The layer height was set at 0.32 mm which is 80% of the orifice and the layer width was set at 0.6 mm which is 150% of the orifice. The bed temperature is kept at a constant 60°C using the default setting and assumption setting in Ultimaker Cura. The printing temperature was tested at various values from 220°C and increased by 20°C until it reached 260°C to find the optimum temperature where increasing from that optimum value would result in no significant difference. The printing speed was also tested at varying velocities starting from 70 mm/s as the initial velocity up until 280 mm/s to determine the optimum speed where increasing from the optimum value would result in no difference in printing time.

The results that were gathered are the experimental flow rate, theoretical flow rate, and the actual speed that the printer is printing at. The theoretical flow rate was calculated by multiplying the layer width, layer height and printing speed used together. The actual printing speed that the Prusa printer was calculated by dividing the experimental flow rate value by the total of layer width and layer height multiplied. The experimental flow rate was calculated by using Equation 7 that was sourced from the Slice Engineering paper [14], with values of the diameter and length of the filament used are 1.75 mm and 1920 mm respectively, with the time value gathered during the printing operation.

$$Q = \frac{\pi d^2 l}{4t} \quad (7)$$

Calculations found that the actual printing speed was found to be significantly lower than the printing speed parameter that was inputted onto Ultimaker Cura, and the optimum speed value is 53.10 mm/s. It also shows that the experimental flow rate was lower than the theoretical flow rate which means that the printer is slower than predicted to achieve optimum results.

It was found that the optimum flow rate for the printer is 10.2 mm³/s achieved using the printing speed of 240 mm/s and 240°C, where the temperature is not further increased as earlier samples show that the product between 240°C and 260°C provides no visual and quality difference along with samples that were printed at 220°C as can be seen in [Appendix E: Physical Testing Analysis](#), where printed products using the varying temperature and speed are located, and the speed is capped at 240 mm/s as faster speed

provides the same printing time as 240mm/s and would, therefore, waste additional energy and materials in its operation, with the actual printing speed of 53.10 mm/s. The printed product using the optimum results and results for every increment for the first method are presented in the respective Figure and Table 5.1.1.



Figure 5.1.1: Printed Product of Method 1 Using Optimum Parameters

Table 5.1.1: Method 1 Results

<i>Sample</i>	<i>Printing Speed (mm/s)</i>	<i>Printing Temperature (°C)</i>	<i>Mass (g)</i>	<i>Theoretical Flow Rate</i>	<i>Experimental Flow Rate (mm³/s)</i>	<i>Actual Print Speed (mm/s)</i>
1	70	220	5	13.4	5.70	29.69
2	70	240	5	13.4	5.77	30.06
3	70	260	5	13.4	5.83	30.37
4	80	220	5	15.4	6.32	32.90
5	80	240	5	15.4	6.33	32.95
6	80	260	5	15.4	6.35	33.09
7	90	220	5	17.3	6.74	35.11
8	90	240	5	17.3	6.77	35.26
9	90	260	5	17.3	6.79	35.37
10	150	240	5	28.8	8.95	46.61
11	200	240	5	38.4	9.60	50.01
12	210	240	5	40.3	9.66	50.32
13	220	240	5	42.2	10.01	52.18
14	225	240	5	43.2	10.11	52.63
15	230	240	5	44.2	10.17	52.98
16	240	240	5	46.1	10.2	53.10
17	250	240	5	48	10.2	53.10
18	280	240	5	53.8	10.2	53.10

5.2 METHOD 2: PRINTING TIME CAPPED AT 600 SECONDS

The second method uses various parameters that were set in Ultimaker Cura, which are layer height, layer width, Bed Temperature, Printing Temperature and Printing Speed. The layer height was set at 0.32 mm which is 80% of the orifice that was used, the layer width was set at 0.6 mm which is 150 % of the orifice that was used. The bed temperature was kept at a constant temperature of 60°C, using the default setting and assumption from Ultimaker Cura. The printing temperature was also kept at a constant of 240°C. The printing temperature is kept at that temperature because method 1 testing showed that there is no significant difference in the printed product between using 240°C and 260°C as the printing temperature. The initial printer speed was set at 70 mm/s and then increased incrementally at a rate of 20 mm/s at each time up to 250 mm/s.

The results that were gathered are the mass, experimental flow rate, theoretical flow rate of the printed object, and the actual printer speed that was used. The theoretical flow rate is calculated by multiplying the layer width, height, and the printing speed used. The experimental flow rate is calculated by using Equation 8 sourced from the Slice Engineering paper [14] with the mass of the printed object values gathered during testing, the density of the PLA filament is 0.00125 mm/g and the time of 600 seconds.

$$Q = \frac{mp}{t} \quad (8)$$

The experimental flow rate calculated shows that it increases every time the printing speed is increased and shows that the mass of the printed product and printing speed are linearly correlated. The actual print speed that Prusa was operating and theoretical flow rate calculation show that the printer is printing at a lower velocity than the velocity value that was inputted into Ultimaker Cura at an optimum speed of 62.5 mm/s and that the flow rate it was operating at was lower than the calculated theoretical flow rate which means that printer is slower than predicted to achieve the optimum results using the set parameters.

The maximum experimental flow rate that was reached 12 mm³/s which is higher than the optimum flow rate from method 1, which means that the optimum flow rate ranges from 10 to 12 mm³/s, which means that the optimum flow rate of Prusa i3 MK3 hotend can be achieved using the parameters of 240 mm/s up to 250 mm/s as the printing speed and 240°C as the printing temperature. The printed optimum sample was printing with the actual printing speed of 62.5 mm/s producing a printed product that weighed 9 grams. The printed samples with varying printing speed settings can be seen in [Appendix E: Physical Testing Analysis](#) and the printed product that was created using the optimum parameters and values and the following results for each increment are presented in the Figure 5.2.1 and Table 5.2.1.



Figure 5.2.1: Printed Product of Method 2 Using the Optimum Parameters

Table 5.2.1: Method 2 Results

Sample	Printing Speed (mm/s)	Printing Temperature (°C)	Mass (g)	Theoretical Flow Rate (mm ³ /s)	Experimental Flow Rate (mm ³ /s)	Actual Print Speed (mm/s)
1	70	240	5	13.44	6.7	34.72
2	90	240	6	17.3	8	41.67
3	110	240	6	21.1	8	41.67
4	130	240	7	24.9	9.33	48.61
5	150	240	7	28.8	9.33	48.61
6	170	240	7	32.6	9.33	48.61
7	190	240	8	36.5	10.7	55.56
8	210	240	8	40.3	10.7	55.56
9	230	240	9	44.2	10.7	55.56
10	250	240	9	48	12	62.5

5.3 THERMAL CAMERA RESULTS

The thermal camera was pointed at the hotend during the printing process using the set parameters that result in the optimum flow rate for the hotend. The following images show the nozzle before its heated up to temperature and the nozzle after its heated up to the set temperature. Based on the second figure, it shows that the heat block and nozzle component experiences the most amount of heat and that the heat sink is experiencing a much lower temperature like its intended to during the process of heating up the hotend.

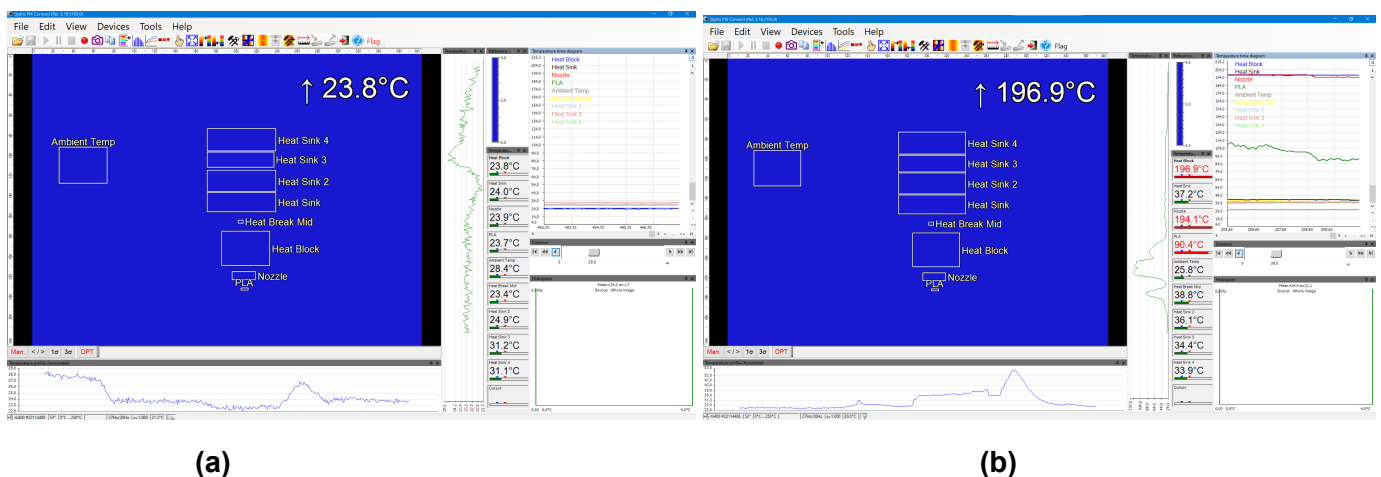


Figure 5.3.1: Hotend Before Heating Up (a) and Hotend After Heating Up (b)

6.0 CREATIVE SOLUTION GENERATION

6.1 DESIGN BOUNDS

The client has requested that improved solutions for their hotend should aim for a 20 to 50% increase in flow rate to accommodate their increasingly large requested prints on their Big Meter Modix printers which measure 1,010 x 1,010 x 1,010 mm. They have also requested that off the shelf nozzle options would be preferred as these parts experience a high frequency of hits whilst in operation and hence need to be replaced regularly. The client has also requested that the designed hotend is compatible with BondTech BMG direct drive extruders.

The slice engineering mosquito has been recommended to use based on experience from the client, thus we will need to incorporate the appropriate mounting points and dimension the heating block to accommodate the mosquito heat break and partnered heat sink to the design. The heating block must also accommodate the addition of a PT1000 thermistor that measures 3x15mm and two 60W heating cartridges which measure 6x50mm each. The list of potential alternatives for the different requirements can be found in the morphology Table at Table 6.6.1.

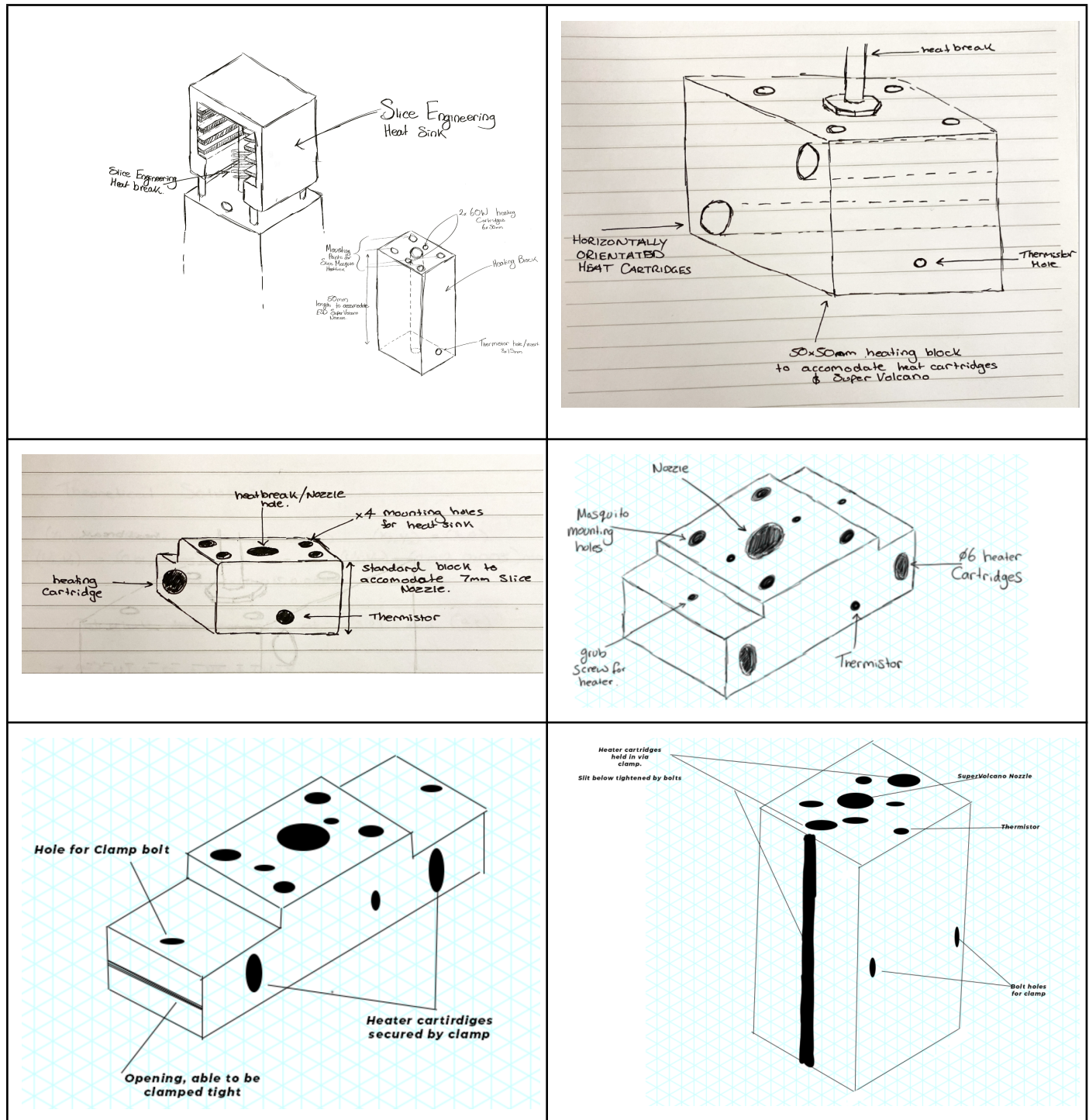
Table 6.1.1: Morphology Table to Generate Different Combinations of Creative Solutions

Morphology Table	Alternatives		
Functions	A	B	C
<i>High Flow Rate</i>	Standard Slice Style (7mm thread length) One of the shortest nozzle sizes that will not increase flow rate any further but will cost less than other off the shelf options	E3D Volcano Nozzle (16mm thread length) A bigger alternative at 16mm thread length allows a greater distance for the filament path to be subjected to a heating process, thus increasing flow rate.	E3D Super Volcano Nozzle (45mm thread length) The longest off the shelf option for a nozzle. With the greatest length of a nozzle, maximises the heating area the filament can undergo and thus ensure the filament to be at a melting temperature
<i>High Fan Cooling Efficiency</i>	GDA010 Standard Slice Heat Break Fan (24V, 0.06A, 11mm deep) Being only 11mm deep allows a lighter and cheaper design. The GDA010 is not the most powerful which makes it best suited for lower melting temperature filaments, ensuring the filament being used isn't being cooled past the heat break into the heating zone	N052AR Heat Break Fan (24V, 0.15A, 28mm deep) A much bigger profile and more expensive, however, has more power than the GDA010 and thus would be able to cool down greater temperatures. Ideal for filaments that have a greater melting temperature	-
<i>Heater Cartridge and Thermistor Orientation</i>	Horizontal heater cartridge and thermistor for a shorter heat block profile	Vertical heater cartridge and thermistor to ensure best possible heat transfer through a taller heat block	Opposing orientations of the heater cartridge and thermistor to satisfy geometry of creative heat block
<i>Mounting Mechanism</i>	Grub screws perpendicular to thermistor or heater cartridge to screw and lock both into the heat block	Open slit with the heater cartridge or thermistor to allow a longer screw to be positioned perpendicular to the slit. Locking in place without contact of the actual cartridge	-
<i>Cost Efficiency for Performance (machining cost, thermal properties etc)</i>	Aluminium 6061 Aluminum has the highest thermal conductivity out of the alternatives and therefore will be able to transfer greater heat from the cartridges into the filament path [19]	Steel Steel has the lowest thermal conductivity from the alternatives and therefore will benefit if the type of filament requires a lower melting temperature to ensure a lower amount of heat is transferred into the filament path [19]	Brass Brass has a thermal conductivity range in between Aluminum and Steel, therefore making it a universal material in usage. However, significantly low or high melting temperature filaments will not be recommended [19]

6.2 GENERATED SOLUTIONS

Utilising Table 6.1.1 with various configurations of alternatives for each function of the hotend, we were able to generate several hotend solutions as shown in Table 6.2.1. These will have differing performance characteristics however we will be able to select two of the best solutions from this process to then compare them in a weighted performance. Table 6.2.1 illustrates several design concepts generated from the varying alternatives derived within Table 6.1.1. Each design tackles functions in differing ways thus, we have a selection to choose from to continue development.

Table 6.2.1: Six Different Potential Design Solutions



6.2.1 CONCEPT 1: SUPERVOLCANO HEATING BLOCK

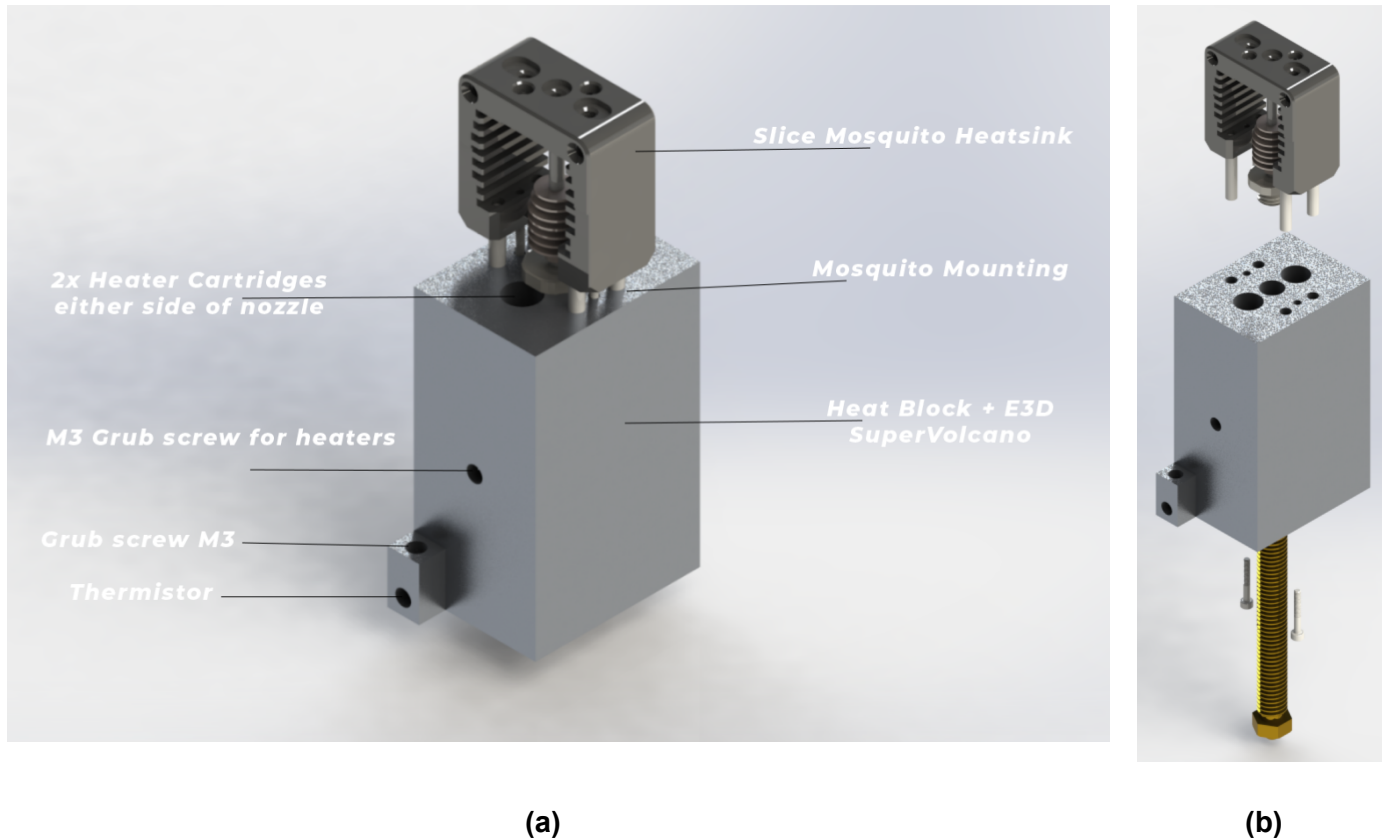


Figure 6.2.1.1: Design Concept 2 with Annotated Isometric View(a) and Exploded View(b)

The main process by which we can increase flow rate to the clients needs would be to maximise the heating time experienced by the filament. From our research, longer nozzles such as the E3D Volcano and Super Volcano are able to achieve greater flow rates to that of shorter competitors as the filament has a longer path to travel and thus more time to be heated by the block. Being a more liquid state it is able to be pushed out faster without jamming. [22]

Concept design 1 integrates the E3D Super Volcano nozzle measuring at $56.5 \pm 0.5\text{mm}$ with the block and nozzle. By designing a heating block around one of the longest available nozzles this concept aims at maximising volumetric flow rate through heating time of filament.⁵ Unlike traditional heatsinks, the Slice engineering heatsink not only attaches to the included heat break but also to the heat block. Therefore, this concept like the others has integrated mounting holes.

Due to the length of the heating block, the heat cartridges have been oriented vertically in the Y direction to maintain constant heating throughout the length of the block. The PT1000 thermistor is able to be inserted into the side of the heating block however due to the requirements for space of other components, an outcrop has been introduced to hold 2/3rds of the length of the thermistor with a hole for a grub screw to be inserted from the top, securing the thermistor as it is not threaded.

6.2.2 CONCEPT 2: STANDARD SLICE NOZZLE HEATING BLOCK

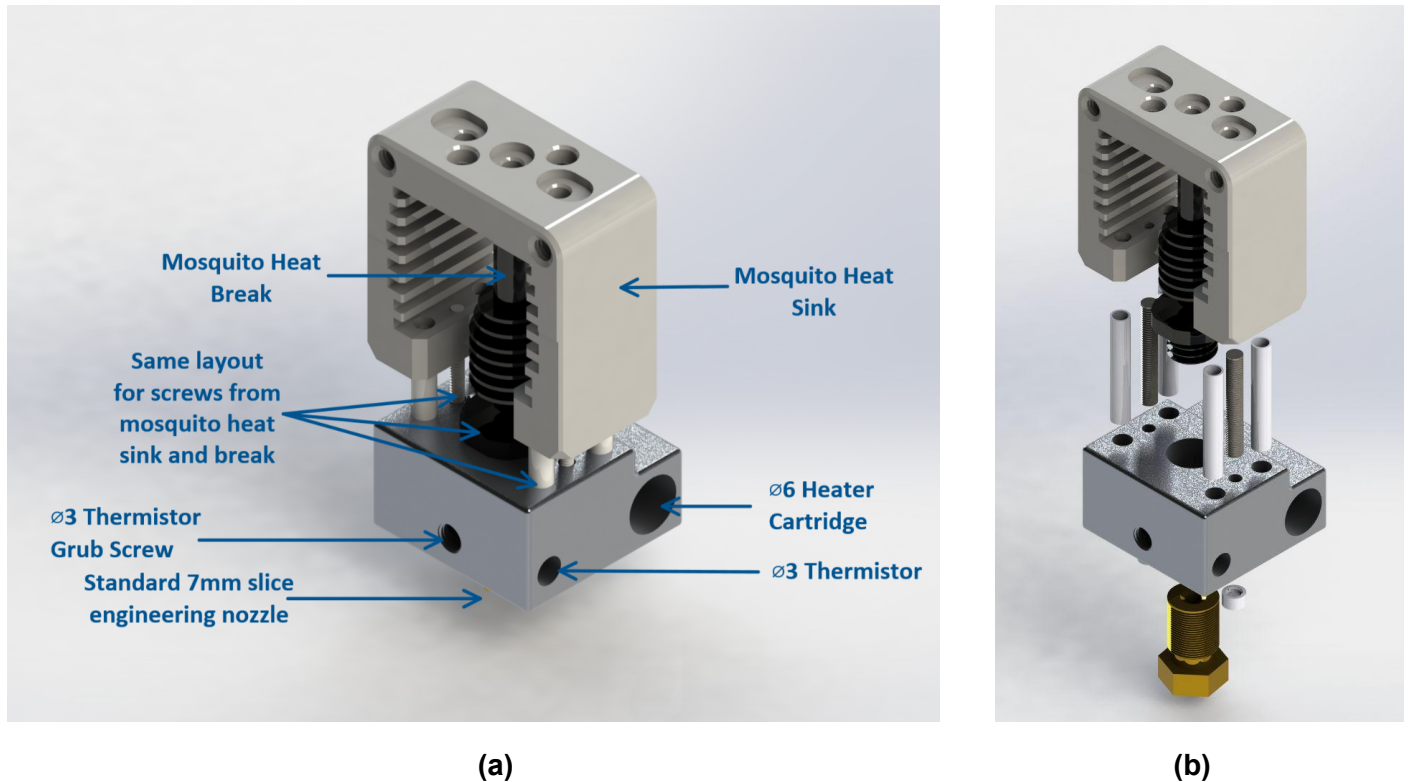


Figure 6.2.2.1: Design Concept 2 with Annotated Isometric View (a) and Exploded View (b)

The main priority for concept design 2 is to minimise cost as much as possible. Taking the opposing focus of having a longer nozzle to maximise the amount of travel distance for the filament to be heated [22], a shorter nozzle allows for a significant difference in weight. When choosing materials, both brass and aluminum alloy are generally cheaper materials in the industry. Therefore, the brass nozzle chosen to be incorporated with the simplistic and prismatic geometry of the heat block made of aluminum alloy, will limit the material cost and machining expense substantially. Essentially sacrificing the possibility of a higher flow rate with a longer nozzle for a cheaper and lighter design.

Visually, design concept 1 is fairly similar to that of the original mosquito hotend design. Implementing a few differences that simplify the heat block design to lower the machining cost and generally a smaller design. Despite the use of a longer nozzle threading length (standard slice length of 7mm), the design concept will weigh lighter than the original mosquito heat block. What is similar between the two heat blocks, is the use of only one heating cartridge. Although two heating cartridges were preferred, the reduction of extra weight and cost is more valuable and significant in the case of the main priority and principles of the concept 2.

Additionally, the main difference between the designs is translating the thermistor's threaded hole perpendicular to the heating cartridge hole. Contrary to the mosquito having the thermistor and heat cartridge holes parallel. The following change in design is to remove any further customization from the underside of the heat block (non-inclusive of the nozzle). Mainly a convenience change so when maintenance is required such as swapping out nozzles, the nozzle is the only swappable component on its surface face. Another change for the mounting mechanism was diverging from any use of a slit when locking in the heater cartridge to the heat block. Machining a slit as a mounting mechanism for the heater cartridge would require more material and surface area on the heat block. Therefore, having a grub screw to lock in the heater cartridge also would allow for an even more minimal design.

Detailed drawings of this concept are available in the [Appendix F: Concept Solutions](#) that illustrates the ease of machining with this design.

7.0 DECISION MAKING

With two concept designs presented, a critical decision making process is needed to select the hotend most suitable for the clients needs. This will be completed by ranking the two concept designs against performance requirements. These performance requirements have individual weightings that will influence the scoring of the concept designs. By ranking each concept against performance criteria we will have a final 'score' that can be used to quantitatively evaluate the parts, and thus select the best one for development. The final decision making criteria and weighting can be seen in Table 7.0.1.

Table 7.0.1: Criteria and Weighting

<i>Criteria</i>	<i>Weighting</i>	<i>Rationale</i>
Volumetric Flow Rate	10	Volumetric flow rate is the main factor in this consulting project. It remains the greatest limiting factor for the client and thus improvements for this metric dominate our weighting.
Affordability	7	Whilst cost was not a limiting factor with the development of the hotend, it was deemed necessary to take it into account and ensure cost to performance ratio is financially acceptable to justify the investment in custom parts.
Strength (structural)	6	FEA analysis highlighted that loads experienced during the routine operation of the printers would not cause serious deformation or damage to the hotend components of the benchmarking printers. An improved hotend should obviously maintain or improve upon this.
Ease of manufacture	5	Manufacturing will affect the cost of the hotend. Designed parts should therefore not have any areas unnecessary for the function of the printer. The more complex the hotend is, the greater the manufacturing cost.
Weight	2	Weight will not be a limiting factor here however should be considered. A hotend with significant weight could negatively affect print speed due to increase load on axis motors and/or result in greater impact loads ($F=m*a$).
Size	2	Size of the hotend will not be a significant issue. However, longer hotends will proportionally decrease the maximum print height within the machine it is being used in. For the client's MODIX Big Meter, this represents a miniscule loss of ~2-4% if the longest nozzle available is used.
Materials (UN SDGs)	4	This criterion specifically references UN SDG's target 9.2 - promoting sustainable industrialisation. The designed hotend should aim to utilise sustainable materials and manufacturing.

Reviewing Table 7.0.1, the construction of Table 7.0.2 allows a quantitative decision to be reached. Table 7.0.2 and Table 7.0.3 will contain scores out of ten from each criteria. The scores are relevant to each of the concepts' ability to satisfy the criterias, accumulating to a total. Therefore, the calculated totals from each concept can be compared and conclude which is the ideal design.

Table 7.0.2: Results from Decision Making of Concept Design 1

Concept 1				
Criteria	Weighting	Non-Dimensional Score (/10)	Reasoning	Score x Weighting
Volumetric Flow Rate	10	10	Integrating two heater cartridges maximises heat production and therefore liquidation of filament. Liquid state of filament greatly affects the rate at which extruders can push in the prints. Largest nozzle, all available heating cartridges will allow for maximum volumetric flow rate.	100
Affordability	7	4	More material and milling required for this hotend, along with an expensive nozzle which will continue cost throughout the heat block lifetime as each nozzle wears and requires replacing.	28
Strength (structural)	6	6	With a longer heat block, hits at the nozzle will generate greater bending moments compared to that of smaller blocks.	36
Ease of manufacture	5	9	Square blocks with reasonable holes and threads. No chamfers or fillets.	45
Weight	2	4	More material, comparatively heavier than baseline Prusa heat blocks and concept 2.	8
Size	2	6	Longer than standard heat blocks, within the MODIX big meter printer, an increase in length from a standard heat block equates to around a 2-4% reduction in print height.	12
Materials (UN SDGs)	4	8	Aluminium being 100% recyclable supports this project's UN SDG goal 9 of sustainable industry. Theoretically a greater flow rate and heating capacity will result in less failed prints, thus less PLA used and thrown out.	32
TOTAL				261/360

Table 7.0.3: Results from Decision Making of Concept Design

Concept 2				
<i>Criteria</i>	<i>Weighting</i>	<i>Non-Dimensional Score (/10)</i>	<i>Reasoning</i>	<i>Score x Weighting</i>
Volumetric Flow Rate	10	5	Standard nozzle length will yield average results	50
Affordability	7	8	Less materials, more affordable nozzle for replacements and uses only one heater cartridge allows for the lowest cost design	56
Strength (structural)	6	9	Smaller build with the same material properties allows for a lower bending moment incurred, thus greater structural integrity throughout the hotend and heat block	54
Ease of manufacture	5	6	A singular block of aluminium alloy with a simple extrusion and screw holes required indicates low cost machining in manufacturing	30
Weight	2	8	Significantly more light weight as the heat block is smaller in size and only one heater cartridge is being used, reducing as much weight as possible	16
Size	2	9	Shorter in all dimensions, leaving a smaller profile. Allowing little to no reduction in the maximum printing height.	18
Materials (UN SDGs)	4	6	Aside from the common recyclability of aluminium itself, each heat block uses a minimal amount of material in manufacturing. Therefore, using less material per printer that will use the smaller design	24
TOTAL				248/360

7.1 DECISION MAKING - STATISTICAL SIGNIFICANCE

Comparing concept 1's score of 261 and concept 2's score of 248, concept 1's bigger design has ranked higher when conducting the Composite Criterion Method. Between the two scores is not significantly different, thus a further analysis and comparison must be made to further differentiate and ensure the right decision is made. Calculating the sensitivity of the decision is completed by taking the largest weighting of criteria (highest weighting value of 10) multiplied by the incremental ranking score (being 0.1). Hence, the sensitivity factor behind the decision was a magnitude of 1. Due to the sensitivity factor being a small value, the gap between concepts 1 and 2 is thirteen multiples of the sensitivity, concluding that concept 1 is the clear choice.

Even disregarding concept 1's composite criterion score, the team chose concept 1 for its superior performance in volumetric flow rate of filament. Relating back to the main objective to improve the design of the hotends attached to the company's Modix 3D printers, increasing the volumetric flow rate as much as possible seemed like the sensible creative goal. Therefore, despite the sacrifices in mainly affordability, weight and size, the significant difference in flow rate that concept 1 can generate is worth it to the team.

Additionally, concept 1 works toward the team's conscious effort to consider our UN goal of sustainable materials and energy usage. Having a greater flow rate allows for less time of energy consumption when the printer is in use. Coupled with the low machining cost and requirement for manufacturing each heat block contributes to lower energy consumption. Contributing to the team's UN goal of sustainable energy usage and consumption in manufacturing. Hence, design 1 is a creative solution of function alternatives incorporating the E3D Super Volcano nozzle, vertical heater cartridge and thermistor orientation, grub screw mounting mechanism and an aluminium build is the chosen design.

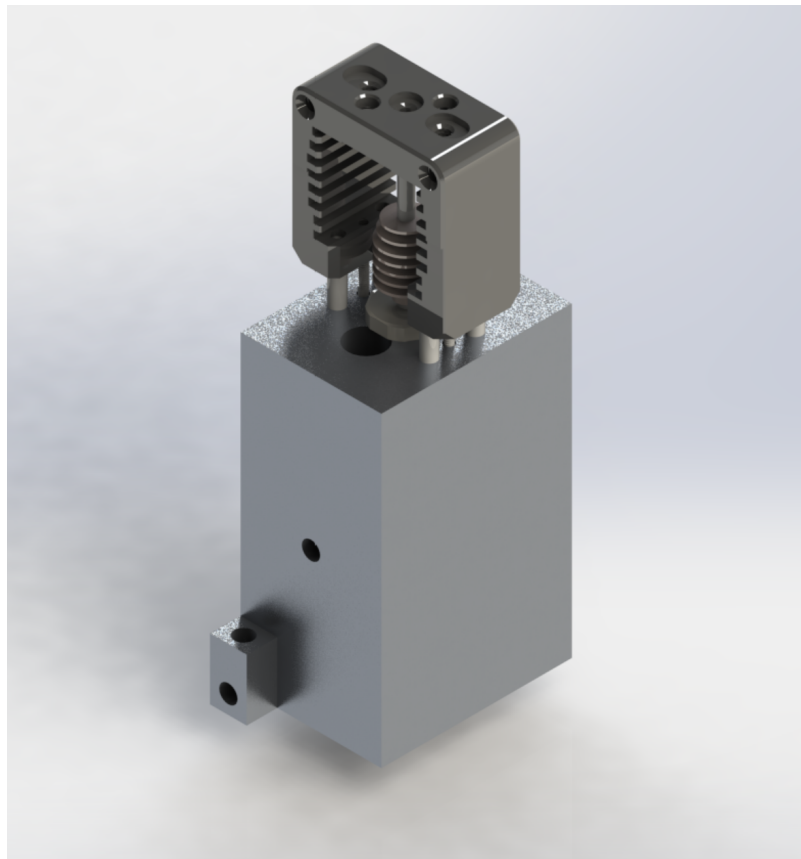


Figure 7.0.4: Final design Concept of the Team's Creative Solution

8.0 PLAN FOR FURTHER WORK

In terms of taking these report findings further, suggesting that the given design concept the team decided on is approved, rigorous testing is mandatory. Once approved, prototyping a model of the design concept is necessary. Benchmarking the prototype would allow the team to validate its estimated capabilities in manufacturing and full industry usage. Optimisation would allow for our final concept to undergo similar benchmarking structurally, thermally and physically to what our team has completed with the E3D V6 hotend. Analysing the structural integrity, thermal efficiency and experimental flow rate of a prototype will reinforce whether or not our design concept is applicable and practical. All leading to a successful outcome of a new and improved hotend design for the client's Modix 3D printers.

However, if the restrictions of this project were to be removed, additional design concepts and more complex creative solutions would be discovered. For instance, the restrictions of nozzle selection severely limits the possibilities of greater flow rates. Specifically when other off the shelf nozzles with greater orifices would be suitable selections for attaining a high flow rate. Given a greater orifice, the filament has a greater exit surface area out of the nozzle, thus generating a greater volumetric flow rate out of the nozzle. Potentially pairing the long nozzle length of our design concept, which allows for a longer distance for the filament to be at a melting temperature, with a greater orifice area at the exit would significantly exceed our concepts' estimated maximum volumetric flow rate.

Additionally, exploring projects outside of just using PLA would expand the creative solutions to include other materials that may be best suited. The team's choice of material for the final concept design is chosen because PLA was the filament being used. However, if future projects require structurally stronger or higher heat resistant filaments to be used, a change in materials is also necessary to adapt. For instance, filament such as polyamide carbon fibre (PAHT CF15) is a high temperature resistant filament used in the automotive industry for engine parts [23]. A greater printing temperature is then necessary to increase the warm up time of the heat block and nozzle. An increase in warm up time can be achieved by swapping out the material of the heat block and nozzle for nickel plated copper for instance [24]. A specific material made for high temperature filaments that can heat the heat block and nozzle to 500°C and allow printers to use and produce parts made from PAHT CF15.

Ultimately, the project objective and outcome is dependent on what creative solution needs to be generated. With every project requiring different materials or different nozzles. There is not a hotend design that is universally fit and suited for any project. It is all dependent on what the client needs.

9.0 PRELIMINARY CONCLUSIONS

In order for the team to create a new solution to improve Modix 3D printing performance, benchmarking is mandatory for helping the team understand what is required for performance increases. CAE testings confirmed the team's initial predictions of the Prusa i3 MK3 printer's E3D V6 hotend design structural and thermal performance. Structurally, rough hand calculations resulted in a force as little as 0.072N in the horizontal plane from a potential collision stopping the nozzle in place. While a force as minimal as 0.0043N was calculated by hand in the z direction to stop the hotend. Both of which are significantly small forces that highlights the impossibility of the hotend failing under physical collisions. Reinforcing the findings from the hand calculations are FEA results using ANSYS. Simulating the same possibility of collisions from all directions, except exaggerating the forces induced by using 40N. Resulting in average safety factors of around 14.7 in the x and z directions, and 14.96 in the y direction before failure of the hotend. Therefore, the structural analysis indicates high structural integrity and strength. Thermally, the team's hand calculations and CAE results reinforced one another and outlined the efficiency of heat transfer across the heat block to the heatsink. Having a temperature of 229.54°C in the heat block and 58.27°C in the heat sink signifies the immense temperature drop in the heat break. Confirming the hotends' ability to prevent heat creep when in use. Physical testing of the E3D V6 hotend illustrated a clear differentiation between the theoretical and experimental flow rate. With the experimental flow rates recorded only reaching up to 12 mm³/s using the second method. Significantly shy of a theoretical flow rate of 48 mm³/s, four times as high. Hence, because of the discrepancy between theoretical and the experimental results in practice allows our team to realise optimisation and prototyping is always required before confirming any theoretical capabilities.

Ultimately, the team has decided on a final design concept that entails our experiments findings. Recommending a heat block and nozzle design that focuses on the client's main priority of maximising flow rate. Therefore, the longest nozzle was chosen as a result of research indicating a longer nozzle ensures a longer distance for the filament to be in a melting state. Ensuring a melting state for the filament allows a smoother exit out the nozzle instead of being extruded while less malleable. Coupled with the simplistic heat block design to limit machining and manufacturing expenses supports the designs affordability in industrial practice. Hence, the final design by the team directly relates to the client's priority of maximising filament flow rate.

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APPENDIX A: CONTRACT OF WORK



Contractual Work Agreement

THIS AGREEMENT is made

BETWEEN:

- (1) **MONASH UNIVERSITY** (ABN 12 377 614 012) a body corporate constituted in accordance with the *Monash University Act 2009* (Vic) of Wellington Road, Clayton, Victoria Australia 3800, acting through its MEC4407 Team 11 (**Supplier**); and
- (2) **3D PRINTING PTY LTD** (ABN 12 377 614 012) of 1 Filament Way, Clayton, Victoria Australia 3800 (**Client**)

BACKGROUND:

- (A) The Client has agreed to provide the Client Contribution (which may include a mix of Fees, Rights, Services and/or Goods) to the Supplier in relation to the Purpose.
- (B) In return for the Client Contribution made by the Client, the Supplier has agreed to grant the Supplier Contribution (which may include a mix of Fees, Rights, Services and/or Goods) to the Client in relation to the Purpose.

IT IS AGREED AS FOLLOWS:

1. TERM

This Agreement commences on the Commencement Date and will continue until the Expiry date, unless terminated earlier in accordance with clause 5.

2. SERVICE

- 2.1 The Client agrees to provide the Client Contribution to the Supplier for the Term as specified in Item 4 of the Schedule for the Term.
- 2.2 The Client represents and warrant that it possesses, or has obtained, the necessary permissions and rights to grant the Supplier the use of the Client Contribution for the Purpose and that the Supplier's use of the Client Support in accordance with this Agreement will not infringe the rights of any third party.
- 2.3 The Supplier agrees to provide the Supplier Contribution to the Client for the Term as specified in Item 5 of the Schedule.

3. INTELLECTUAL PROPERTY

- 3.1 Each party agrees that nothing in this Agreement transfers or shall be deemed to transfer ownership of any of its Intellectual Property rights to the other party.
- 3.2 For the purposes of this clause "Intellectual Property Rights" means all intellectual property rights including without limitation patents, trade marks, service marks, trade and service names, copyrights, database rights and design rights (whether or not any of them are registered and including applications for registration of any of them), rights in know-how, moral rights, trade secrets and rights of confidence, all rights or forms of protection of a similar nature or having similar or equivalent effect to any of them which may subsist anywhere in the world at the date of this Agreement or in the future.

4. GST

- 4.1 Words and expressions in this clause that are defined in the *A New Tax System (Goods and Services Tax) Act 1999* (Cth) have the meanings given them in that Act.
- 4.2 If a party makes a supply under or in connection with this Agreement in respect of which GST is payable, and the consideration does not already include GST, the consideration for the supply (but for the application of this clause) ("GST exclusive consideration") is increased by an amount equal to the GST exclusive consideration multiplied by the rate of GST. For the avoidance of doubt, where the supply is for non-monetary consideration, the GST inclusive market value of the consideration provided is used to determine the price and value of the supply.
- 4.3 If a party must reimburse or indemnify another party for a loss, cost or expense, the amount to be reimbursed or indemnified is first reduced by an amount equal to any input tax credit the other party is entitled to in respect of the loss, cost or expense, and then increased in accordance with this GST clause (if applicable).
- 4.4 Each party agrees that if it makes a taxable supply under or in connection with this Agreement, it will provide a tax invoice for that supply in accordance with the provisions of the Act.
- 4.5 If at any time an adjustment of GST arises in connection with any supply made or deemed to be made under or in connection with this Agreement by the supplier, a corresponding adjustment to the GST amount must be made between the supplier and the recipient and any payment necessary to give effect to the adjustment must be made. In addition, the supplier must provide an Adjustment Note to the recipient at the same time the adjustment occurs.

5. TERMINATION

- 5.1 Either party may terminate this Agreement at any time during the Term by giving to the other party one (1) month notice in writing.
- 5.2 The Supplier may immediately terminate this Agreement in its absolute discretion by giving written notice to the Client if:
- (a) a conflict of interest exists;
 - (b) the Client's conduct may, in the reasonable opinion of the Supplier, bring the Supplier, or the Purpose, into disrepute;
 - (c) the Client repudiates this Agreement;
 - (d) the Client ceases to carry on business;
 - (e) any step is taken to enter into any arrangement between the Client and its creditors;
 - (f) any step is taken to appoint a receiver, a receiver and manager, a trustee in bankruptcy, a liquidator, a provisional liquidator, an administrator or other like person of the whole or any part of the Client's assets or business.
- 5.3 Upon the expiry or termination of this Agreement:
- (a) the respective obligations of each party in relation to the Client Contribution due after the date of expiry or termination will cease;
 - (b) each party will cease to have any Rights; and
 - (c) any Contribution granted to a party under this Agreement will terminate.
- 5.4 Expiry or termination of this Agreement does not affect accrued rights and remedies.

6. DISPUTE RESOLUTION

- 6.1 The parties will attempt to resolve any dispute arising under this Agreement by negotiation in good faith, initiated by one party giving written notice of the dispute to the other party. If a dispute remains unresolved within thirty (30) days of a party receiving written notice of the dispute, the parties will refer the dispute to mediation. The mediator will be appointed jointly by the parties or, where the parties cannot agree, nominated by the then President of the Law Institute of Victoria or if not then in operation, a like organisation to be agreed between the parties. All costs of the mediation shall be borne equally by the parties, and the parties shall bear their own legal costs. In the event the dispute is not resolved by mediation within twenty (20) days after the appointment of the mediator, or such further time as the parties agree in writing, a party may then commence proceedings in relation to the dispute.
- 6.2 Notwithstanding the existence of a dispute, the parties must continue to perform their obligations under this Agreement.
- 6.3 Nothing in this clause 6 shall be taken as preventing a party from commencing a proceeding for urgent interlocutory relief or from terminating this Agreement in accordance with clause 5.

7. FORCE MAJEURE

- 7.1 Neither party will be liable for any delay or failure to perform its obligations under this Agreement where such delay or failure is due to any event beyond that party's reasonable control. The parties agree to use reasonable endeavours to remove or overcome the effects of the relevant event as soon as practically possible.
- 7.2 Should an event in connection with this Agreement be postponed or cancelled in connection with an event contemplated by clause 7.1, the Supplier agrees to refund any Fees paid by the Client and the parties each agree to release the other from any further obligation under this Agreement.

8. GENERAL

- 8.1 This Agreement is governed by and must be construed in accordance with the laws of Victoria, Australia. The parties submit to the non-exclusive jurisdiction of the courts of Victoria.
- 8.2 This Agreement may only be amended or varied by written agreement of the parties.
- 8.3 The Client must not assign or otherwise transfer its rights or obligations under this Agreement without the prior written consent of Supplier.
- 8.4 The parties are independent contracting parties, and nothing in this Agreement shall create a Suppliership, joint venture or agency relationship between the parties, nor does it grant either party any authority to assume or to create any obligation on behalf of or in the name of the other.
- 8.5 This Agreement constitutes the entire agreement and understanding between the parties with respect to the subject matter of this Agreement.
- 8.6 If a term of this Agreement is invalid or unenforceable it shall be severed to the extent of the invalidity or unenforceability, without affecting the remaining terms.
- 8.7 Each party agrees that this Agreement may be executed by electronic signature (regardless of the form of electronic signature utilised) and that this method of signature is conclusive of the parties' intention to be bound by this Agreement as if physical signing had occurred.
- 8.8 This Agreement may be executed in any number of counterparts and by the parties on separate counterparts. Each counterpart constitutes the agreement of each party who has executed and delivered that counterpart. Each party may communicate its execution of this Agreement by successfully transmitting an executed copy of the Agreement by an electronic method to each party.

9. DEFINITIONS

9.1 The following additional definitions apply in this Agreement

Agreement means this agreement between the parties and includes the Schedule.

Item means an item listed in the Schedule to this Agreement.

Fees means financial payments granted by the Supplier to the Client, or the Client to the Supplier, in relation to the Purpose as set out in Item 4 and/or 5 of the Schedule.

Rights means the rights and benefits granted by the Supplier to the Client, or the Client to the Supplier, in relation to the Purpose as set out in Item 4 and/or 5 of the Schedule.

Services means the action of helping or the provision of work, granted by the Supplier to the Client, or the Client to the Supplier, in relation to the Purpose as set out in Item 4 and/or 5 of the Schedule.

Goods means both tangible and intangible materials and products, granted by the Supplier to the Client, or the Client to the Supplier, in relation to the Purpose as set out in Item 4 and/or 5 of the Schedule.

Client Contribution means the mix of Fees, Rights, Services and/or Goods provided by the Client to the Supplier in relation to the Purpose as set out in Item 4 of the Schedule.

Supplier Contribution means the mix of Fees, Rights, Services and/or Goods provided by the Supplier to the Client in relation to the Purpose as set out in Item 5 of the Schedule.

Term means the start and finish date of this Agreement set out in clause 1.

Purpose can be provided as either: the general mission or objective of the Supplier organisation; or the specific mission or objective of the Clientship Agreement made with the Supplier as set out in Item 1 of the Schedule.

SCHEDULE

1.	Purpose	The purpose of this engagement is to create a high-performance 3D printer hotend design that exceeds the capabilities of an existing consumer-grade Slice Mosquito hotend by 20% to 50%. This enhancement aims to enable faster and more reliable 3D printing for the Client's large-scale projects. Upon completion, the project should deliver an innovative, efficient, and structurally sound hotend design that improves flow rate, thermal performance, and overall printing quality. This design will be both developed and manufactured, and its performance will be thoroughly validated for the Client's use.
2.	Commencement Date	Wednesday 30 August 2023
3.	Expiry Date	Wednesday 18 October 2023
4.	Client Contribution	The Client is expected to commit at least one hour every three weeks to meet with the Supplier to check on the progress and provide input and feedback. The Client is also expected to respond to any emails or calls regarding the project from the Supplier within 2 business days. The total cost of the project will be AUD\$353,332.43 (GST inclusive). This total cost includes labor costs, material costs, overhead costs, technology and tools costs, and other miscellaneous costs, all required by the Supplier to fulfill the contractual needs. A more detailed breakdown of the quote is available in Attachment 1: Costing Quote. Upon execution of this contract, an initial payment of AUD\$100,952.12 (GST inclusive) shall be made, payable to the Supplier within 2 days of the signing date. Subsequent progress payments of AUD\$100,952.12 (GST inclusive) shall be made fortnightly on 13 September 2023 and 27 September 2023, as mutually agreed upon by both parties, and payable to the Supplier within 3 business days of the invoice date from the Supplier. The final payment of AUD\$50,476.06 (inclusive of GST) shall be made within 10 days of the project's completion and the submission of the final deliverable, the presentation, by the Supplier.

5.	Supplier Contribution	The Supplier is expected to dedicate at least one hour every three weeks to meet with the Client, providing updates on the project's progress and remaining open to feedback. Additionally, the Supplier is required to respond to any emails or calls regarding the project from the Client within 2 business days. Throughout the 7-week duration, the Supplier should engage in an interactive prototyping process, which involves finalising a design, manufacturing it, conducting theoretical and physical testing, and subsequently optimising the design. This process is expected to occur at least twice during the project timeline. Upon project completion, the Supplier should have developed a 3D printer hotend that aligns with the Client's specifications and the purpose of this contract. To validate the design's viability, the Supplier is responsible for conducting thermal, structural, and physical testing on the prototype. As the project draws to a close, the Supplier is required to compile a comprehensive report summarising the project's work and outcomes. Additionally, the Supplier should deliver a presentation to the Client, showcasing the details of their work and the results achieved. While the Supplier has the option to provide the Client with a physical final solution, this is not an obligation. The Supplier will retain the Intellectual Property rights related to the hotend's design upon project completion and will grant the Client permission to manufacture the product for installment and use on their own Modix Printers.
6.	Approval & Record Keeping	For this contract to be considered valid, a draft of this Agreement must be submitted to the Supplier's Authorised Officer for a comprehensive review and subsequent approval. Upon obtaining the required approval and the subsequent signatures from both parties, a digital copy of the fully executed Agreement must be forwarded via email to the Supplier's Authorised Officer and the Supplier's Senior Supervisor. Failure to fulfill either of these conditions may result in the contract becoming void at the discretion of the Supplier.

EXECUTED as an agreement.

signed for and on behalf of MONASH UNIVERSITY by its authorised officer:		
Print name		Signature
Title		Date

signed for and on behalf of 3D PRINTING PTY LTD by its duly authorised officer:		
Print name		Signature
Title		Date

Signed and acknowledged by the Supplier's Senior Supervisor:		
Print name		Signature
Title		Date

ATTACHMENT 1: COSTING QUOTE

MONASH UNIVERSITY

Wellington Road, Clayton, Victoria Australia 3800

<https://www.monash.edu/>

+61 3 9902 6000

ABN 12 377 614 012



MONASH University

BILL TO

QUOTE DATE: 25 August 2023

VALID FOR: 14 days

Scott Wordley

3D PRINTING PTY LTD

ABN 12 377 614 012

1 Filament Way, Clayton, Victoria Australia 3800

DESCRIPTION	QUANTITY	UNIT PRICE	TOTAL
<u>LABOUR COST</u>			
Engineering Graduates 40 hours/week for 7 weeks (including Superannuation)	5	\$63.08 /hr	\$88,312.00
Senior Engineer Supervisor 20 hours/week for 7 weeks (including Superannuation)	1	\$209.05 / hr	\$29,267.00
Machinist 5 hours/heat block for 2 heat block iterations (including Superannuation)	1	\$ 52.46 / hr	\$524.60
TOTAL LABOUR COST			\$118,103.60
<u>MATERIAL COST</u>			
Slice Mosquito Heat Sink	1 (+1 Spare)	\$63.00	\$126.00
Slice Mosquito Heat Break - 1.75 mm	1 (+1 Spare)	\$64.00	\$128.00
Slice Mosquito Heat Block Hardware including Standoff tubes, Bolts and other Bolts not included by Slice	1 (+1 Spare)	\$15.00	\$30.00
60 W Heater Cartridge	2 (+1 Spare)	\$55.00	\$165.00
N052AR Heat Break Fan	1 (+1 Spare)	\$20.00	\$40.00
PT1000 Thermistor	1 (+1 Spare)	\$30.00	\$120.00
Brass E3D Super Volcano Nozzle	1 (+1 Spare)	\$16.00	\$32.00
Aluminium Alloy 6060 Round Bar, 50mm Diameter 1m Length	1	\$147.53	\$147.53
PLA 3D Filament 1kg Roll	2 (+1 spare)	\$34.95	\$104.85

SUB TOTAL MATERIAL COST			\$893.38
SHIPPING COST PERCENTAGE ESTIMATE			30%
SHIPPING COST ESTIMATE			\$268.01
TOTAL MATERIAL COST			\$1,161.39
<u>OVERHEAD COST</u>			
Office Space for the whole team for the 7 weeks	1	\$37,589.04	\$37,589.04
Utilities for the whole team for the 7 weeks including Electricity and Gas, Internet, Water and Office Kitchen	1	\$16,233.58	\$16,233.58
Administration Expenses for the whole team for the 7 weeks such as paying wages and salaries	1	\$20,997.23	\$20,997.23
Security Expenses for the whole team for the 7 weeks	1	\$37,800.00	\$37,800.00
Maintenance and Cleaning Experiences for the whole team for the 7 weeks	1	\$9,800.00	\$9,800.00
TOTAL OVERHEAD COST			\$122,419.85
<u>TECHNOLOGY AND TOOLS COST</u>			
ANSYS Software Licence for the Whole Team for the 7 Weeks for the following licences: Ansys Mechanical, Ansys Mechanical Solver, Ansys Structural, Ansys Structural Solver, Ansys Mechanical CFD-Flo	1	\$17,076.17	\$17,076.17
SolidWorks Premium Software Licence for the Whole Team for the 7 Weeks	1	\$8,046.74	\$8,046.74
Fusion360 Software Licence for the Whole Team for the 7 Weeks	1	\$735.00	\$735.00
MATLAB Software Licence for the Whole Team for the 7 Weeks	1	\$1,095.46	\$1,095.46
Microsoft 365 Business Premium Software Licence for the Whole Team for the 7 Weeks	1	\$317.10	\$317.10
Zoom Business Plus for the Whole Team for the 7 Weeks	1	\$201.37	\$201.37
TOTAL TECHNOLOGY AND TOOLS COST			\$27,471.84
<u>MISCELLANEOUS COSTS</u>			
Intellectual Property - Design Rights Registration	1	\$450.00	\$450.00
Business Insurance Costs	1	\$262.50	\$262.50

Legal Costs	1	\$1,925.00	\$1,925.00
MISCELLANEOUS COST			\$2,637.50
SUB TOTAL			\$271,794.18
TAX RATE			30%
TOTAL TAX			\$81,538.25
SHIPPING AND HANDLING			\$0.00
QUOTE TOTAL (INCLUDING GST)			\$353,332.43

APPENDIX B: STRUCTURAL HAND CALCULATION ANALYSIS

To calculate bending moments incurred during a possible collision of the E3D V6 hotend, a summation of forces including a hit and inertia aided to calculate a final bending moment that can be found in table B.2. In addition, thread stripping forces were calculated to ensure its impossibility of failure in table B.2.

Table B.1: Results for Calculating Directional Hits

	<i>Maximum Velocity (mm/s)</i>	<i>Maximum Acceleration (mm/s²)</i>	<i>Weight (N)</i>	<i>Inertia (N)</i>	<i>Impulse (kgm/s)</i>	<i>Directional Hit (Collision) (N)</i>
<i>X/Y Direction</i>	200	1000	N/A	0.036	0.0072	0.072
<i>Z Direction</i>	12	200	0.35	0.0072	0.00043	0.0043

Table B.2: Results for Calculating Bending Moment

	<i>Tensile Stresses (Pa)</i>	<i>Compressive Stresses (N)</i>	<i>Shear Area On Threaded Screws (m²)</i>	<i>Thread Stripping (kN)</i>	<i>Maximum Support Reaction Forces Incurred (N)</i>	<i>Maximum Bending Moment Incurred (Nm)</i>
<i>X/Y Direction</i>	80.32	N/A	N/A	N/A	0.11	0
<i>Z Direction</i>	N/A	41.0	0.00013	18.6	0.36	0.0069

APPENDIX C: STRUCTURAL CAE ANALYSIS

The different contour plots for the x, y and z hit conditions can be seen in Figure C.1, Figure C.2 and Figure C.3, respectively.

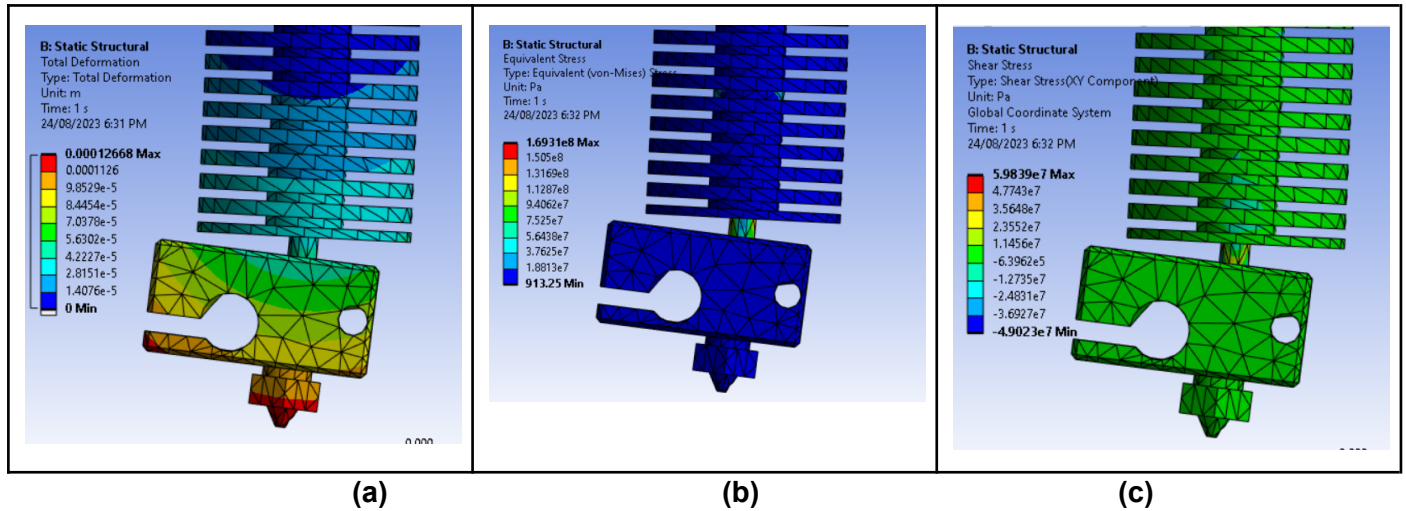


Figure C.1: Deformation (a), Stress (b) and Shear (c) contour of X hit 40N

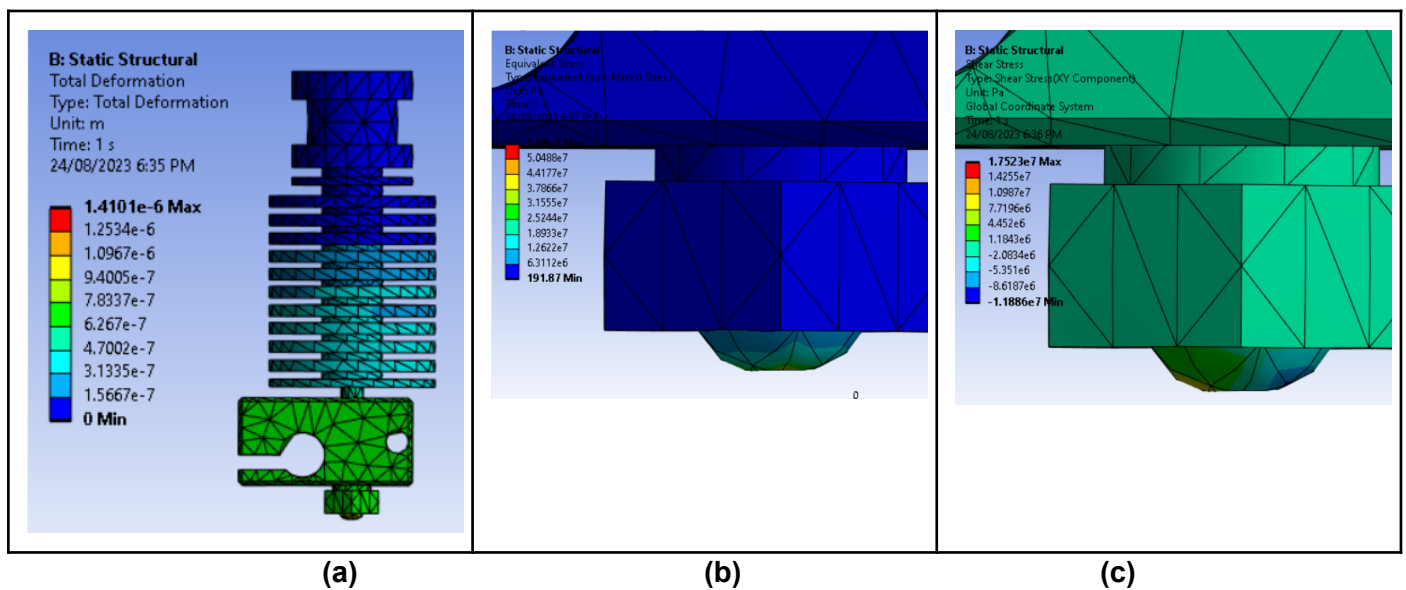


Figure C.2: Total Deformation (a) Stress (b) and Shear (c) Contour Y hit at 40N

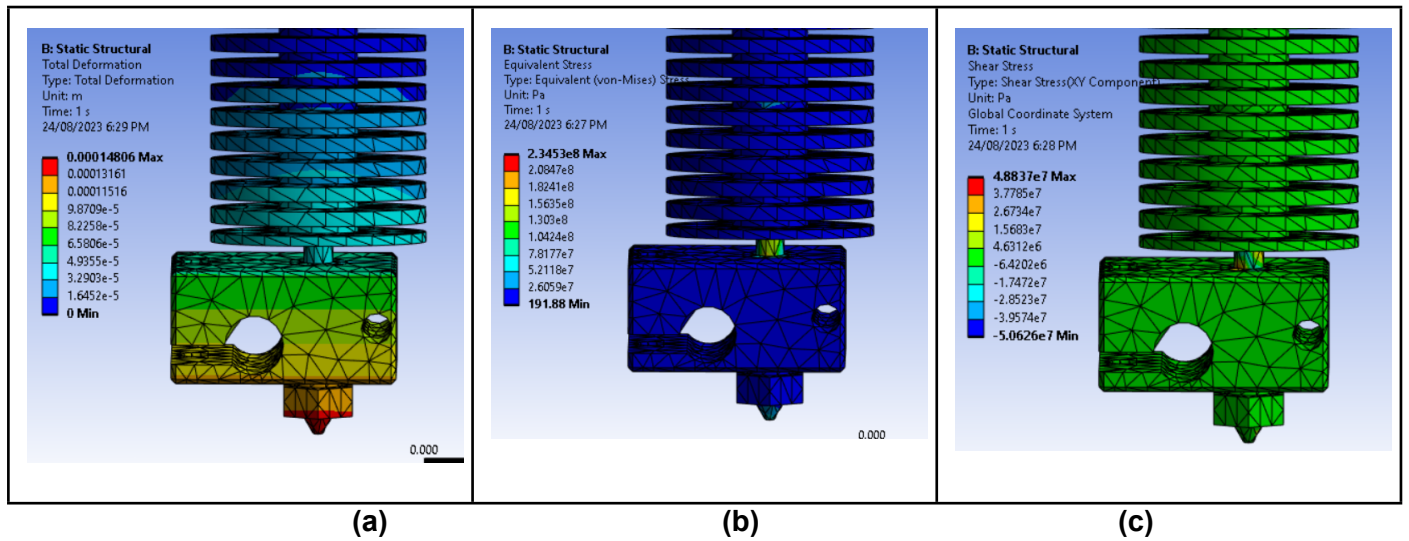


Figure C.3: Deformation (a) Stress (b) Shear (c) Z hit at 40N

Additionally, a summary of the relevant values generated by ANSYS can be seen in Table C.1.

Table C.1: ANSYS Values

Object Name	Disco\E3D_v6_Heat break-1		Disco\E3D_V6_Heater block-1		Disco\E3D_v6_nozzle0.4mm-1		Disco\E3D_v6_Heat sink-1	
State	Meshed							
Graphics Properties								
Visible	Yes							
Transparency	1							
Definition								
Suppressed	No							
Stiffness Behavior	Flexible							
Coordinate System	Default Coordinate System							
Reference Temperature	By Environment							
Treatment	None							
Material								
Assignment	Stainless steel, 316, annealed		Aluminum alloy, wrought, 6061, T6		Brass, C37700		Aluminum alloy, wrought, 6061, T6	
Nonlinear Effects	Yes							

Thermal Strain Effects	Yes							
Bounding Box								
Length X	7.e-003 m		2.3074e-002 m		7.e-003 m		2.2e-002 m	
Length Y	2.21e-002 m		1.15e-002 m		1.25e-002 m		4.27e-002 m	
Length Z	7.e-003 m		1.6109e-002 m		8.0829e-003 m		2.2e-002 m	
Properties								
Volume	5.5423e-007 m³		3.0277e-006 m³		2.9784e-007 m³		7.7158e-006 m³	
Mass	4.4166e-003 kg		8.2142e-003 kg		2.4623e-003 kg		2.0933e-002 kg	
Centroid X	5.3364e-003 m		2.1543e-003 m		5.3365e-003 m			
Centroid Y	1.5962e-002 m		4.8901e-003 m		-6.5698e-004 m		3.0984e-002 m	
Centroid Z	4.5161e-002 m		4.5145e-002 m		4.5161e-002 m			
Moment of Inertia Ip1	1.5867e-007 kg·m²		2.9445e-007 kg·m²		3.2866e-008 kg·m²		3.3894e-006 kg·m²	
Moment of Inertia Ip2	2.7475e-008 kg·m²		5.7923e-007 kg·m²		1.4044e-008 kg·m²		1.1587e-006 kg·m²	
Moment of Inertia Ip3	1.5867e-007 kg·m²		4.8181e-007 kg·m²		3.2866e-008 kg·m²		3.3894e-006 kg·m²	
Statistics								
Nodes	1362		3962		1286		6177	
Elements	702		2023		646		2827	
Mesh Metric	None							
CAD Attributes								
PartTolerance:	0.00000001							
Color:175.143.175								
Color:159.175.143								
Color:143.143.175								
Color:175.159.143								

APPENDIX D: THERMAL ANALYSIS

To develop a better understanding of a current competitor's hotend, thermal CAE simulation results were analysed for the Prusa i3 MK3 printer's E3D V6 hotend design. The objective of the thermal analysis is to assess the performance of the E3D V6 heat break and calculate the maximum flow rate of the hotend.

It is important to assess the performance of the heat break as it will indicate whether heat creep occurs. PLA has a glass temperature (temperature at which it starts to soften) of around 60°C and a melting temperature of around 180°C [6]. For a well designed and effective hotend, the glass temperature should only truly begin at the bottom of the heat sink and melting should occur throughout the heat block. However, sometimes due to poor engineering design, poor cooling systems or overly heating, heat creep can occur. Heat creep is when filament melts earlier than required due to heat unintentionally travelling up the hotend [25]. This is not only bad for the printer but it can drastically affect the quality of the print or even lead it to potentially failing. The heat break serves as the channel for this heat to travel upwards and, therefore, making it essential to analyse it.

It is also important to analyse the maximum potential flow rate of the hotend to determine if the flowrate is limited by the power or whether it is due to the design of the hotend. Heat is inputted through a heater cartridge and diffused through the top and bottom of the hotend as well as through the filament. A poorly designed hotend could have extensive amounts of heat resistance throughout the hotend which results in not enough power being provided to heat up the filament itself. Hence, analysing the maximum potential flow rate is essential to seeing whether the hotend is power limited or whether there are design changes that need to be made to enhance the flow rate.

EQUIPMENT

The two main parts of the Prusa i3 MK3 printer that were required for this analysis include the E3D V6 hotend and the Prusa Noctua Hotend Fan - both of which are the typical components in a Prusa i3 MK3. The Prusa Noctua Hotend Fan, seen in Figure D.1, is the stock standard fan used in Prusa i3 MK3 printers, with a reported maximum fan airflow of 8.2 mm³/hr and a fan diameter of 37 mm. Meanwhile, the typical E3D V6 hotend, as can be seen in Figure D.2, consists of many components (11 different types).



Figure D.1: Prusa i3 MK3 Noctua Hotend Fan [26]

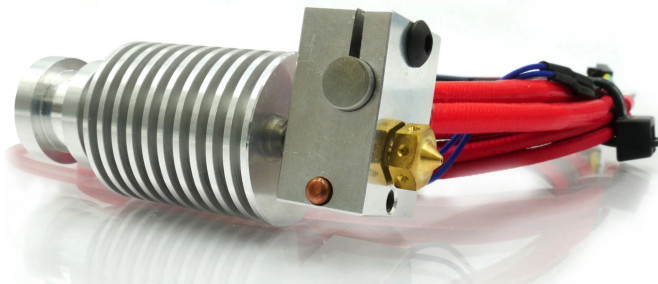


Figure D.2: E3D V6 Hotend [27]

For modelling, only the four major components of the hotend were considered including the nozzle, the heat block, the heat break and the heatsink, as can be seen in Figure D.3.

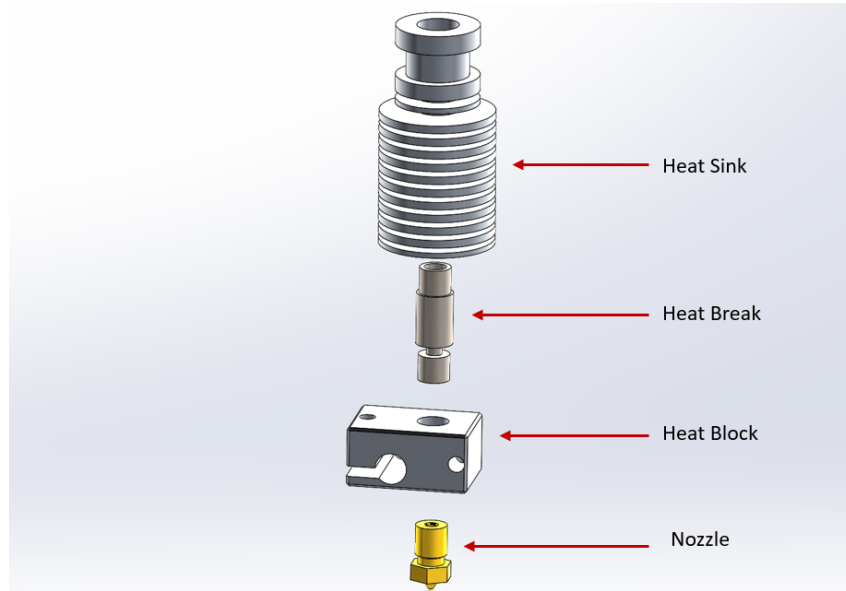


Figure D.3: E3D V6 Hotend Essential Parts Exploded View

MATERIALS

Before performing any calculations or simulations, it is important to determine the necessary material properties required. For the four essential parts of the hotend, the thermal conductivity was determined by extracting the values from ANSYS GRANTA Materials Data for Simulation table (in ANSYS WorkBench) and the values are summarised in Table D.1. It is important to note that these values are at a temperature of 300 K.

Table D.1: Material Properties of the Four Major Components of the Hotend at 300 K [21]

<i>Component</i>	<i>Material</i>	<i>Type</i>	<i>Thermal Conductivity (W/mK)</i>
Nozzle	Brass	C37700	112.1
Heat Block	Aluminium Alloy	6061, T6	155.3
Heat break	Stainless steel	316	14.58
Heatsink	Aluminium Alloy, 6061	6061, T6	155.3

Additionally, the material properties of PLA will be essential to determining the flowrate and its properties at 300 K can be seen in Table D.2.

Table D.2: Material Properties of PLA

<i>PLA Properties at 300 K</i>	
<i>Density (kg/m³)</i>	1240 [28]
<i>Specific Heat Capacity (kJ/kg K)</i>	1800 [29]

Furthermore, using Equation 9, the film temperature was calculated to be 399 K.

$$T_{film} = \frac{T_{\infty} + T_s}{2} \quad (9)$$

This temperature was used to interpolate the material properties of air from a properties table [30] and the final values can be seen in Table D.3.

Table D.3: Material Properties of Air at the Film Temperature extracted from [30]

<i>Air Properties at 399 K</i>	
<i>Kinematic Viscosity (m²/s)</i>	0.0000263002
<i>Thermal Conductivity (W/mK)</i>	0.033724
<i>Density (kg/m³)</i>	0.873578
<i>Specific Heat Coefficient (kJ/kg K)</i>	1.0139
<i>Dynamic viscosity (kg/ms)</i>	0.0000229662

TEMPERATURE SETTINGS AND CONDITIONS

For both the hand and CAE analysis, the input temperature is set to 230°C while the ambient temperature is set to 22°C.

For the hand calculations, a summary of the necessary temperature settings and conditions can be seen in Figure D.4. The temperature is inputted at the center of the heater cartridge and is applied throughout the whole heat block (red). Natural convection is expected to occur with the ambient air of 22°C at the end of the nozzle and the top of the heat block (green), while forced convection with the ambient air of 22°C occurs at the top of the heat sink (purple).

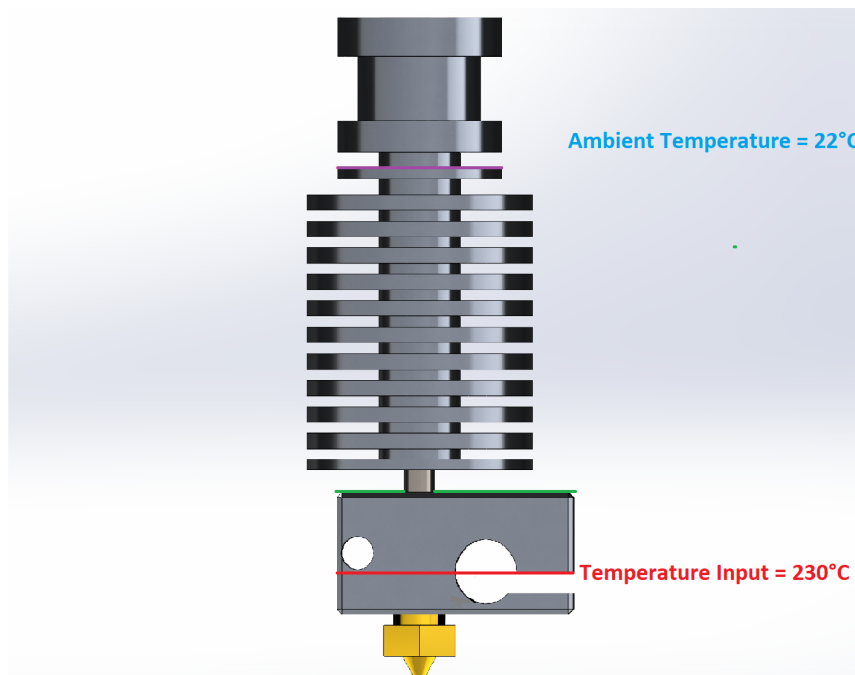


Figure D.4: Temperature Settings and Conditions for Hand Calculations

For the CAE analysis, a summary of the necessary temperature settings and conditions can be seen in Figure D.5. Forced convection occurs throughout the whole heat sink (Figure D.5(a)), while natural convection occurs with the ambient air of 22°C along the bottom half of the hotend (Figure D.5(b)). The constant initial temperature is inputted along the faces of which the heat cartridge touches (Figure D.5(c)).

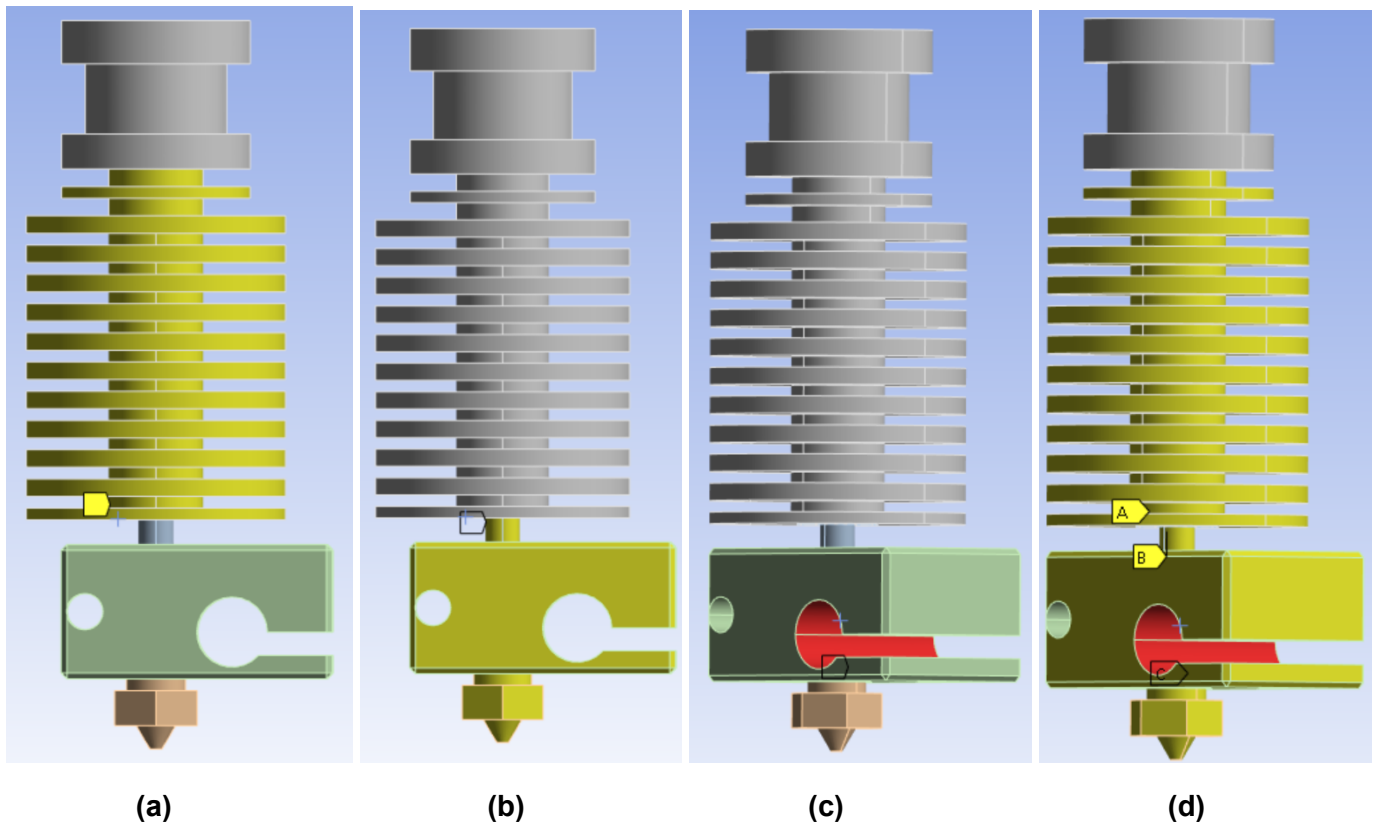


Figure D.5: Temperature Settings and Conditions for CAE Analysis.

Figure D.5(a) shows the forced convection condition.

Figure D.5(b) shows the natural convection condition.

Figure D.5(c) shows the temperature input condition.

Figure D.5(d) shows all of the conditions in one.

ASSUMPTIONS

In order to perform analysis, assumptions had to be made to simplify the complexity of analysis.

For the hand calculations, the necessary assumptions included:

1. The heat transfer is in steady state.
2. There is only heat transfer in one direction - the z-axis direction. Meaning that the temperature is the same in the other directions. This also means that heat only transfers up and down and it cannot leave sideways.
3. The thermal material properties mentioned are constant, whereby any change due to temperature is negligible.
4. Only the four major components are necessary to modelling the heat transfer of the hotend.
5. There is negligible contact resistance between the different parts that are in contact with each other.
6. There are negligible radiation effects.
7. The nozzle, heat break and heat sink can be modelled as a hollow cylinder with different outer radius depending on location. While the heat block can be modelled as a rectangular prism with a hole for the nozzle and heat break.
8. There is negligible natural convection everywhere except for the bottom of the nozzle and the top of the heat block. Where natural convection is considered, the air is assumed to be quiescent.
9. The resistance through the neck of the heat break can be accurately modelled by discretizing it and splitting it up 5 times.
10. The heat break component internal of the heatsink extends all the way up to fin number 7 of the heat sink.

11. The ambient temperature is constant at 22°C (295 K). Meanwhile, the material properties of the forced air is considered at a film temperature which is the average between the ambient air and the maximum temperature of the hotend.
12. The maximum airflow rate of the fan is the actual airflow rate.
13. The PLA has laminar flow.
14. The heat input is considered as a temperature and this temperature is considered to be uniform and located at the center distance of the cartridge, neglecting the circular nature of the heat cartridge.
15. Any material that overlaps is the same temperature.
16. The very top fin has the same dimensions as the rest of the fins.

For the CAE analysis, similar assumptions include assumptions 3, 4, 6, 7, 9, 12, 13. Other necessary assumptions for the ANSYS model included:

1. The heat transfer coefficient for natural convection is constant throughout the whole hotend, nozzle and neck and heat break.
2. The heat in is modelled as a constant inputted temperature that touches only the curved surfaces that the heat cartridge would normally touch.
3. The threaded parts have had their threading removed.

HAND CALCULATIONS

CONVECTION HEAT TRANSFER COEFFICIENTS

To determine the convection heat transfer coefficient for forced convection, it is necessary to determine the speed of the air. A rearrangement of the continuity equation, outlined in Equation 10, was used to determine the speed of the air.

$$\begin{aligned}
 Q_1 &= Q_2 & (10) \\
 Q_1 &= V_2 A_2 \\
 V_2 &= \frac{Q_1}{A_2}
 \end{aligned}$$

The maximum flow rate possible by the fan is known and the area could be calculated by summing up the area of each of the gaps, as outlined in Figure D.6. This generated a flow rate along the fins of ~14.79 mm³/s.

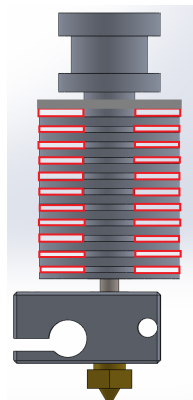


Figure D.6: The Red Boxes indicate the area that is summed to calculate the speed along the fins.

Prandtl's number (Equation 11) and Reynold's number (Equation 12) were determined in order to determine the appropriate Nusselt equation to use (Equation 13), from which the convection heat transfer coefficient can be calculated using Equation 14.

$$Pr = \frac{\nu}{\alpha} \quad (11)$$

$$Re_D = \frac{\rho U D}{\mu} \quad (12)$$

$$\bar{Nu}_D = 0.3 + [0.62 Re_D^{1/2} Pr^{1/3} \times [1 + (0.4/Pr)^{2/3}]^{-1/4}] \times [1 + (Re_D/282,000)^{5/8}]^{4/5} \quad (13)$$

for Cylinder Geometry, using T_{film} and satisfying $Re_D \times Pr \geq 0.2$

$$\bar{h} = \frac{\bar{Nu}_D k}{D} \quad (14)$$

Ultimately, the forced convection heat transfer coefficient can be calculated to be 54.43 W/mK.

For natural convection, the heat transfer coefficient at the bottom of the nozzle and the top of the hotend can be calculated using Prandtl's number (Equation 11), Volume expansion coefficient (Equation 15) and Grashof's number (Equation 16) to determine the Rayleigh's number (Equation 17), necessary in determining the correct Nusselt equation, which ended up being Churchill and Chu's correlation for vertical plates for convection at the top of the hotend (Equation 18) and Churchill and Chu's correlation for long horizontal cylinders for convection at the bottom of the nozzle (Equation 19), from which the convection heat transfer coefficient could be calculated using Equation w.

$$\beta = \frac{1}{T_{film}} \quad (15)$$

$$Gr_L = \frac{g \beta (T_s - T_\infty) L_c^3}{\nu^2} \quad (16)$$

$$Ra_L = Gr_L Pr \quad (17)$$

$$\bar{Nu}_L = 0.68 + \frac{0.670 Ra_L^{1/4}}{[1 + (0.492/Pr)^{9/16}]^{4/9}}, \text{ for } Ra_D \leq 10^9 \quad (18)$$

$$\bar{Nu}_L = \{0.60 + \frac{0.387 Ra_D^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}}\}^2, \text{ for } Ra_D \leq 10^{12} \quad (19)$$

Ultimately, these yielded a natural convection heat transfer coefficient at the bottom of the nozzle and at the top of the heat block to be 5.52 W/mK and 4.50 W/mK, respectively.

RESISTANCE, HEAT OUTPUT AND TEMPERATURE DISTRIBUTION

Given that the convection heat transfer coefficients were now known and the thermal conductivity values were also provided, the resistance could be calculated. Equation 20 was used to calculate the resistance of the conduction through the heat block, while Equation 21 was used to calculate the resistance of the conduction through the nozzle, heat break and hotend.

$$R_{cond, plate} = \frac{L}{kA} \quad (20)$$

$$R_{cond, cylinder} = \frac{\ln(r_2/r_1)}{2\pi Lk} \quad (21)$$

Meanwhile, the convection, both natural or forced, can be calculated using Equation 22.

$$R_{conv} = \frac{1}{hA} \quad (22)$$

Where heat transfer overlapped due to parts overlapping or there being air inbetween parts, the equivalent parallel resistance was calculated using Equation 23.

$$R_{eq, parallel} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \dots \right)^{-1} \quad (23)$$

Meanwhile, as heat flowed in one direction, often the equivalent series resistance was needed to be calculated, which was performed using Equation 24.

$$R_{eq, series} = R_1 + R_2 + \dots \quad (24)$$

It is important to note that the neck of the heat break was discretized into 5 pieces, while the heat sink (and parts of the heat break inside the heat sink) was discretized into 24 pieces, to account for all the gaps and fins. Once the total resistance was known, the heat that exited through the bottom and through the top could be calculated using Equation 25.

$$Q = \frac{\Delta T}{R} \quad (25)$$

Given that the heat that leaves the bottom and top of the hotend can be calculated, this means that all of the temperatures along all of the paths can also be calculated as the heat transfer is in one dimension. Overall, the energy heat balance of the system is depicted in Equation 26.

$$Q_{in} = Q_{out, bottom} + Q_{out, top} + Q_{out, filament} \quad (26)$$

The heat energy required for the filament can be vary, depending on Q_{in} . As such, Equation 27 can be used to calculate the mass flow rate, needed to determine the volumetric flow rate of the PLA done using Equation 28 - as the density of PLA is known.

$$Q_{out, filament} = \dot{m} c \Delta T$$

$$\dot{m} = \frac{Q_{out, filament}}{c \Delta T} \quad (27)$$

$$\dot{q} = \frac{\dot{m}}{\rho} \quad (28)$$

THERMAL CAE ANALYSIS

For the thermal CAE Analysis, the convective heat transfer coefficient terms as well as the input temperature and ambient temperature were entered on the respective faces shown in Figure D.5(d). The appropriate materials were also assigned. Finally, the meshing was checked and different meshes were tried. Ultimately, the mesh shown in Figure D.7, seemed to be the most appropriate, as the temperature values converged while maintaining a low operation time.

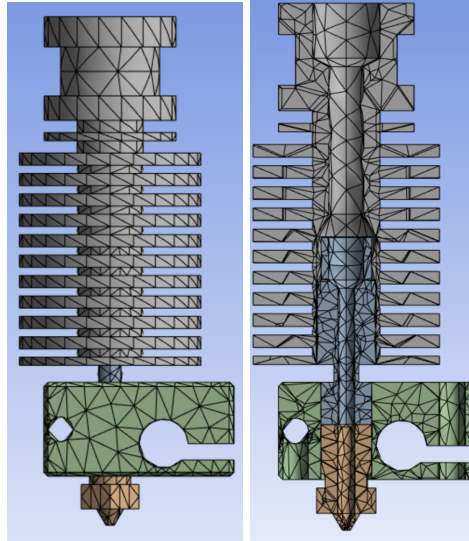


Figure D.7: Front View and Section view of the final mesh used.

HAND CALCULATIONS SPREADSHEET

Hand calculations were performed using a Google Spreadsheet. This spreadsheet can be accessed and viewed using the following link:

<https://docs.google.com/spreadsheets/d/1fsVkCIG-t0dGwVyQlwXVW5KmR9o-RsWSl6Z1d0ueKSs/edit?usp=sharing>

Note that the first sheet contains the hand calculations for the heat transfer coefficient, while the second sheet contains the hand calculations for the resistance and total heat lost through the top and bottom. Additionally, note that any cell that is green is an input cell, whereby changing that value will change appropriate cells in white.

APPENDIX E: PHYSICAL TESTING ANALYSIS



Figure E.1: Sample 9 Method 1 Printed Model at 260°C



Figure E.2: Sample 8 Method 1 Printed Model at 240°C



Figure E.3: Sample 7 Method 1 Printed Model at 220°C



Figure E.4: Sample 1 Method 2 Printed Model at 70 mm/s



Figure E.5: Sample 10 Method 2 Printed Model at 250 mm/s



Figure E.6: Sample 1 Method 1 Printed Model at 220°C and 70 mm/s



Figure E.7 : Sample 9 Method 2 Printed Model at 210 mm/s

APPENDIX F: CONCEPT SOLUTIONS

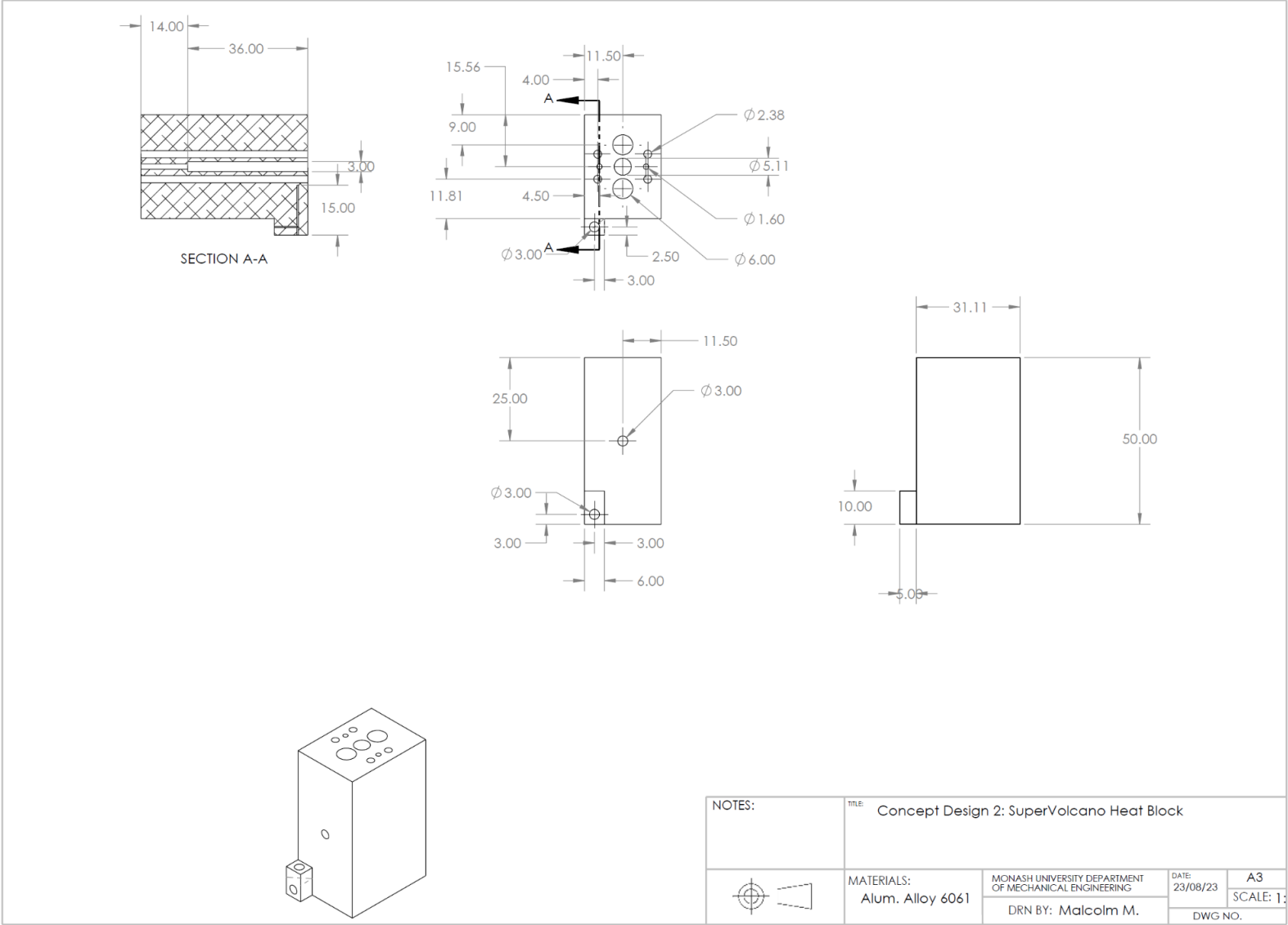


Figure F.1: Concept 1: SuperVolcano Heat Block

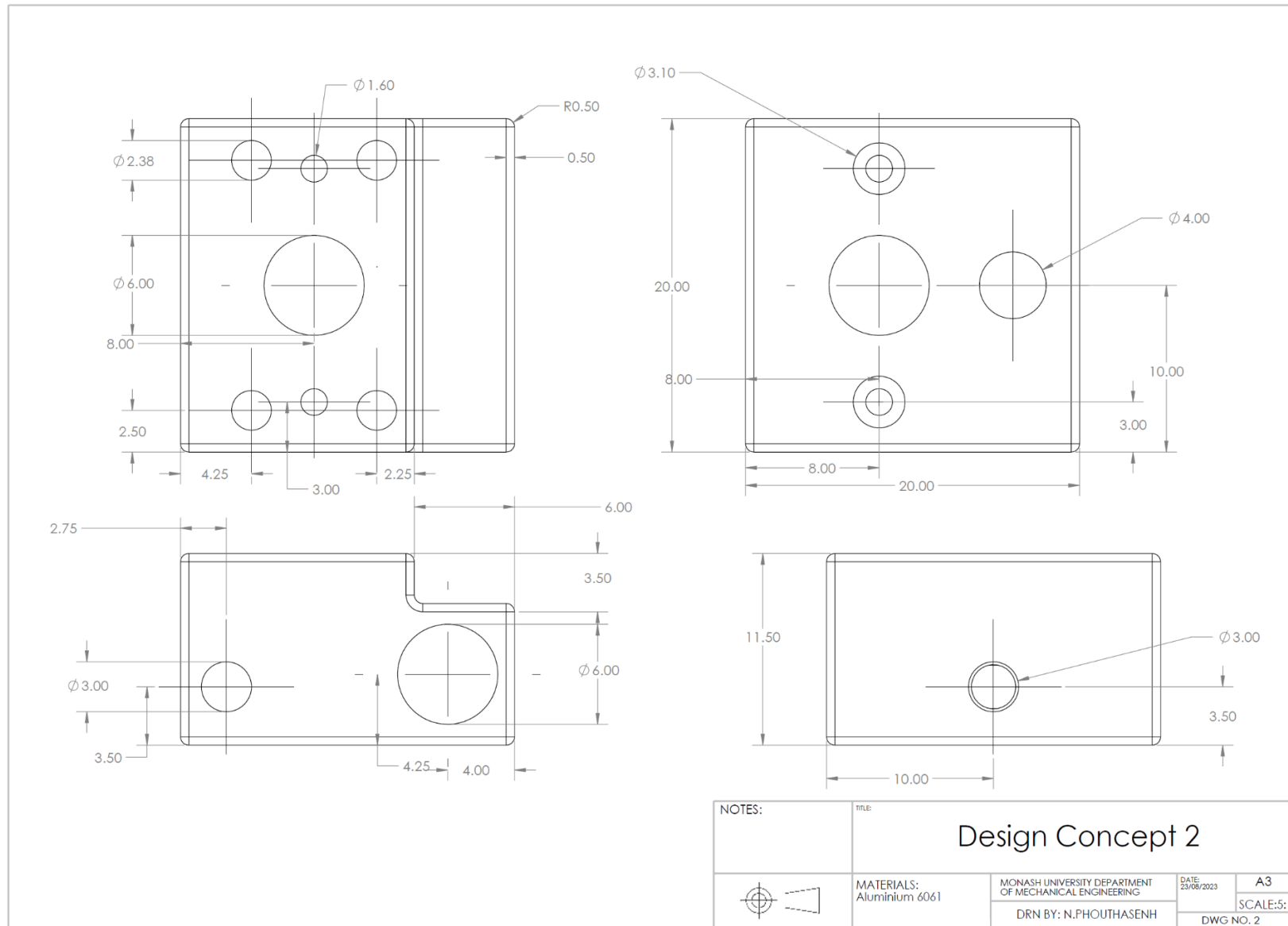


Figure F.2: Concept 2: Standard Slice Engineering Nozzle Heat Block